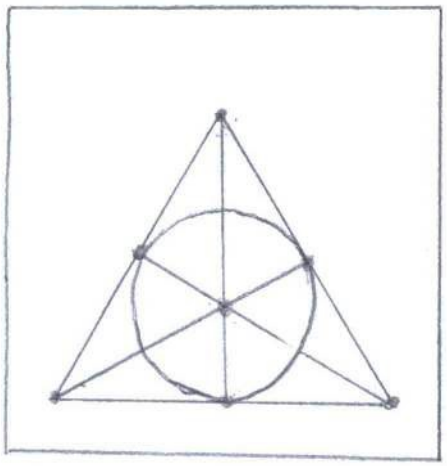
Chapter 9: Geometric Transformations

Exercises

1.



"seven-point projection"

Collineations:

$$\binom{7}{6}\binom{6}{5}\binom{4}{3} = \frac{7!}{6!}\frac{6!}{5!}\frac{4!}{3!} = 168 \text{ collineations}$$

- $\binom{7}{6}$: seven total points
- $\binom{6}{5}$: six points leftover
- $\binom{4}{3}$: collineation of four points, no three collinear

40% correct!

2.

Groups:

#1: $S, T \in G \Rightarrow ST \in G$

#2: $I \in G$

#3: $T \in G \Rightarrow T^{-1} \in G$

#4: $S(TU) = (ST)U$ for all $S, T, U \in G$

(from exercise) "The set of all automorphisms of a model of neutral geometry is a group."

Proof	Hypothesis	Group Property

Incidence:

Axiom I-1:

A) Points P and Q form line $\overleftrightarrow{l}$ — #1, #2

B) $l \xrightarrow{Rot(R)=Rot(R^{-1})} l'$ — #3

C) Points P' and Q' on line $\overleftrightarrow{l'}$ — #4

Axiom I-2:

A) Line $\overleftrightarrow{l}$ with points P and Q — #1, #2

B) $\overline{PQ} \xrightarrow{Rot(l)=Rot(l')} \overline{P'Q'}$ — #3

C) Line $\overleftrightarrow{l'}$ — #4

Axiom I-3:

A) Points P, Q, R are not collinear — #1, #2

B) $P, Q, R \xrightarrow{Rot(R)=Rot(R^{-1})} P', Q', R'$ — #3

C) No collinear line — #4

Betweeness:

Axiom B-1:

A) $A * B * C$ — #1, #2

Points A, B, C are collinear $A*B*C$ and $C*B*A$.

B) $A*B*C \xrightarrow{Rot(R)=Rot(R^{-1})} A'*B'*C'$ #3

C) $C'*B'*A'$ #4

Axiom B-2:

A) Points B, D have other points on $\overleftrightarrow{BD}$: A, C, E. So that $A*B*D$, $B*C*D$ and $B*D*E$ #1, #2

B) $B, D \xrightarrow{Rot(R)=Rot(R^{-1})} B', D'$ #3

C) A', C', E' on $\overleftrightarrow{B'D'}$ #4

Axiom B-3:

A) A, B, and C are on a line, with only one point between the other two. #1, #2

B) $A, B, C \xrightarrow{Rot(R)=Rot(R^{-1})} A', B', C'$ #3

C) A line $A*B*C$ where only one is between the other two. | #3, #4

Axiom B-4:

A) For every line $\overleftrightarrow{l}$ and A, B, C not lying on $\overleftrightarrow{l}$ | #1

i) If A and B are on the same side of $\overleftrightarrow{l}$, and if B and C are on the same side of $\overleftrightarrow{l}$, then A and C are on the same side of $\overleftrightarrow{l}$

ii) If A and B are on opposite sides of l, and if B and C are on

opposite sides of $\overleftrightarrow{l}$, then A and C are on the same side of $\overleftrightarrow{l}$

B) $\overleftrightarrow{l} \xrightarrow{Rot(R)=Rot(R^{-1})} \overleftrightarrow{l}$ #3

C) For every line $\overleftrightarrow{l'}$ and for any three points A', B', and C' not lying on $\overleftrightarrow{l'}$ #4

i) If A' and B' are on the same side of $\overleftrightarrow{l'}$ and if B' and C' are on the same side of $\overleftrightarrow{l'}$, then A' and C' are on the same side of $\overleftrightarrow{l'}$.

ii) If A' and B' are on opposite sides of $\overleftrightarrow{l'}$ and if B' and C' are on

opposite sides of ℓ', then A' and C' are on the same side of $\overleftrightarrow{\ell'}$.

Axiom C-1:

A) A and B are unique points, and, A' and B', where $\overline{AB} \cong \overline{A'B'}$ — #1, #2

B) $\overline{A'B'} \xrightarrow{Rot(R)=Rot(R^{-1})} \overline{A''B''}$ — #3

C) $\overline{AB} \cong \overline{A''B''}$ — #4

Axiom C-2:

A) $\overline{AB} \cong \overline{CD}$ and $\overline{AB} \cong \overline{EF}$, so $\overline{CD} \cong \overline{EF}$ — #1, #2

B) $\overline{AB} \xrightarrow{Rot(R)=Rot(R^{-1})} \overline{A'B'}$ — #3

C) $\overline{A'B'} \cong \overline{CD}$ and $\overline{A'B'} \cong \overline{EF}$ — #4

Axiom C-3:

A) $A*B*C$ IF, $A'*B'*C'$ — #1, #2

$\overline{AB} \cong \overline{A'B'}$ and
$\overline{BC} \cong \overline{B'C'}$ then
$\overline{AC} \cong \overline{A'C'}$

B) $\overline{A'C'} \xrightarrow{Rot(R)=Rot(R^{-1})} \overline{A''C''}$ | #3

c) $\overline{AB} \cong \overline{A''B''}$ and
$\overline{BC} \cong \overline{B''C''}$ | #4.

Axiom C-4:

A) $\angle BAC \cong \angle B'A'C'$ | #1, #2

B) $\angle B'A'C' \xrightarrow{Rot(R)=Rot(R^{-1})} \angle B''A''C''$ | #3

c) $\angle BAC \cong \angle B''A''C''$ | #4

Axiom C-5:

A) $\angle A \cong \angle B$ and $\angle A \cong \angle C$ | #1, #2
then $\angle B \cong \angle C$

B) $\angle A \xrightarrow{Rot(R)=Rot(R^{-1})} \angle A'$ | #3

c) $\angle A' \cong \angle B$ and | #4
$\angle A' \cong \angle C$, so $\angle B \cong \angle C$

Axiom C-6(SAS):

A) $\triangle ABC \cong \triangle DEF$ | #1, #2

when a side, angle, and side are congruent.	
B) $\triangle ABC \xrightarrow{Rot(R)=Rot(R^{-1})} \triangle A'B'C'$	#3
C) $\triangle A'B'C' \cong \triangle DEF$	#4

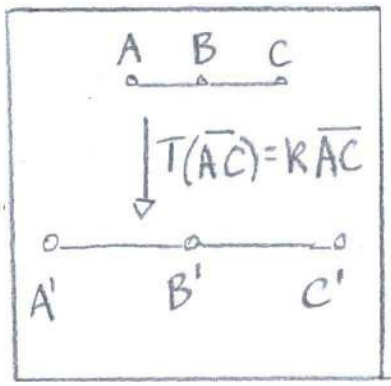
3. (pg 409) Proposition 9.2.

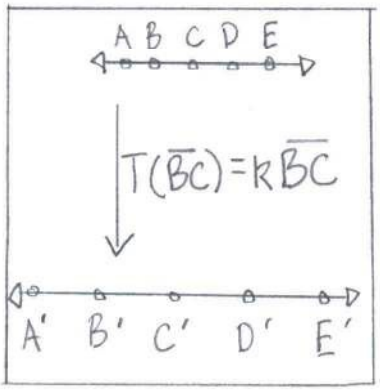
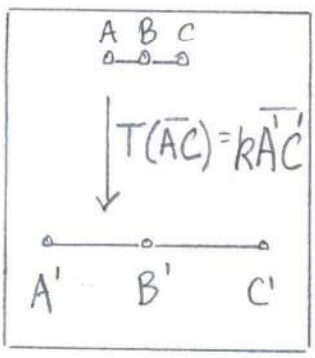
"If T is an automorphism of an Archimedean Hilbert plane, then $\measuredangle A \cong \measuredangle A'$ for every angle. Thus T preserves angle measures."

Proof: Hypothesis	Evidence
1) Transformation T is a motion	(pg 408) "Definition - T"
2) Angle $\measuredangle A$	(pg 13) "Definition - angle"
3) A transformation upon $\measuredangle A$ happens by $T(\measuredangle A) = \measuredangle A'$	Proposition 4.1a
4) $\measuredangle A$ has a total	Measurement Theorem 4.3

degree $(\measuredangle A)^\circ$.	
5) Angle $\measuredangle A'$ has a total degree $(\measuredangle A')^\circ$.	Measurement Theorem 4.3
6) $(\measuredangle A)^\circ = (\measuredangle A')^\circ$ if and only if $\measuredangle A = \measuredangle A'$	Measurement Theorem 4.3

4. (from exercise) "A transformation of a plane that multiplies all lengths by a constant $K>0$ preserves betweeness and congruence of segments."

Proof: Hypothesis	Evidence
Axiom B-1:	
	
A) Points A, B, C are in a plane	(pg XIX "Definition-plane"
B) Points A, B, and C are collinear, A*B*C	(pg 70) "Definition-collinear
C) Transformation (T) multiplies lengths by a constant $K>0$. $T(\text{segment}) = KA'C'$	(pg) "Definition - T"

D) $T(\overline{AC}) = k\overline{A'C'}$	Proposition 9.1
E) $C * B * A$ because $\overline{AC} = \overline{AC} \cup \overline{CA}$	(pg 45) "Definition-Union"
Axiom B-2:	
 A) $\overline{BD}$; From points B and D; also has points A, C, and E.	Axiom B-2
B) A transformation (T) scales length	(pg 132) "Definition-length"
C) $T(BD) = kB'D'$	Proposition 9.1
D) $A' * B' * D'$, $B' * C' * D'$, and $B' * D' * E'$	(pg 70) "Definition-collinear"
Axiom B-3:	
 A) A, B, and C are three distinct points on the same line	(pg 12) "Definition-Points"
B) $T(\overline{AC}) = k\overline{A'C'}$	Proposition 9.1
C) Only B' is between A' and C'	Axioms B-1, B-2

Axiom B-4 (Plane Separation):

A) $\overleftrightarrow{l}$ be a line, then A, B, and C any points not on $\overleftrightarrow{l}$.

(pg12.) "Definition- lines, points"

B) Transformation T scales length

(pg 408) "Definition- Transformation

C) $T(\overline{AC}) = k\overline{A'C'}$ and $T(\overline{AB}) = k\overline{A'B'}$ and $T(\overline{BC}) = k\overline{B'C'}$

Proposition 9.1

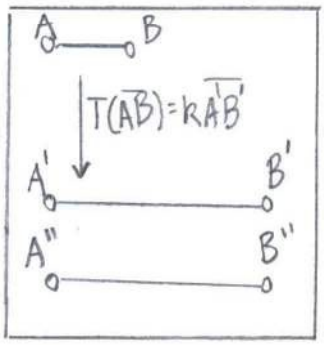
i) If A' and B' are on the same side of $\overleftrightarrow{l}$ and if B' and C' are on the same side of $\overleftrightarrow{l}$, then A' and C' are on the same side of $\overleftrightarrow{l}$.

(pg 110) "Definition- same side"

ii) If A' and B' are on opposite sides of $\overleftrightarrow{l}$ and if B' and C' are on the same side of $\overleftrightarrow{l}$, then A' and C' are on the same side of $\overleftrightarrow{l}$.

(Pg 110) "Definition- opposite sides"

Axiom C-1:



A) If A and B are distinct points with a third point A"

(Pg 12) "Definition- Points"

B) Transformation T scales length

(Pg 403) "Definition- Transformation"

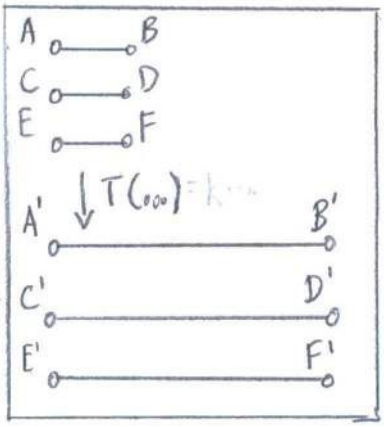
C) $T(\overline{AB}) = k\overline{A'B'}$

Proposition 9.1

D) There is a point B" such $B" \neq A"$ and $\overline{A'B'} = \overline{A"B"}$

Euclids Postulate II

Axiom C-2:



A) If $\overline{AB} \cong \overline{CD}$ and $\overline{AB} \cong \overline{EF}$

(pg 16) "Definition - segment"

B) Transformation T Scales segments

(pg 408) "Definition - Transformation

C) $T(\overline{AB}) \cong k\overline{A'B'}$

$T(\overline{CD}) \cong k\overline{C'D'}$

$T(\overline{EF}) \cong k\overline{E'F'}$

Proposition 9.1

D) $\overline{C'D'} \cong \overline{E'F'}$

Measurement Theorem 4.3

Axiom C-3:

A) If $A*B*C$, $A'*B'*C'$, then $\overline{AB} \cong \overline{A'B'}$, and $\overline{BC} \cong \overline{B'C'}$

(pg 70, 16) "Definitions - collinear, segments"

B) Transformation T Scales lengths

(pg 132) "Definition - lengths"

C) $T(\overline{A''B''}) = k\overline{A''B''}$, $T(\overline{A'''B'''}) = k\overline{A'''B'''}$

Proposition 9.1

$T(\overline{BC}) = k\overline{B''C''}$

$T(\overline{B'C'}) = k\overline{B'''C'''}$

C) $\overline{A''C''} = \overline{A'''C'''}$

Measurement Theorem 4.3

Axiom C-4:

A) Angle $\angle BAC$

(pg 18) "Definition-angle"

B) Ray $\overrightarrow{A'B'}$

(pg 108) "Definition-ray"

C) Transformation T scales length

(pg 408) "Definition-Transformation"

D) $T(\overrightarrow{A'B'}) = k\overrightarrow{A''B''}$, such $\angle BAC \cong \angle B''A''C$

(pg 408) "Definition-motion"

Axiom C-5:

A) $\angle A \cong \angle B$ and $\angle A \cong \angle C$

(pg18) "Definition-angle"

B) Transformation T scales angle

(pg408) "Definition-Transformation"

C) $T(\angle A)^\circ = k(\angle A)^\circ$

Proposition 9.1

$T(\measuredangle B)^\circ = k(\measuredangle B')^\circ$

$T(\measuredangle C)^\circ = k(\measuredangle C)^\circ$

D) $(\measuredangle B)^\circ \cong (\measuredangle C)^\circ$ | Measurement Theorem 4.3

Axiom C-6 (S. A. S.):

A) $\triangle ABC \cong \triangle DEF$ by $\overline{AB} \cong \overline{DE}$, $\measuredangle A \cong \measuredangle D$, and $\overline{AC} \cong \overline{DF}$ | RAA hypothesis

B) Transformation T scales sides | (pg 18) "Definition- sides"

C) $T(\overline{AB}) = k\overline{A'B'}$

$T(\overline{DE}) = k\overline{D'E'}$

$T(\overline{AC}) = k\overline{A'C'}$

$T(\overline{DF}) = k\overline{D'F'}$

$T(\overline{BC}) = k\overline{B'C'}$

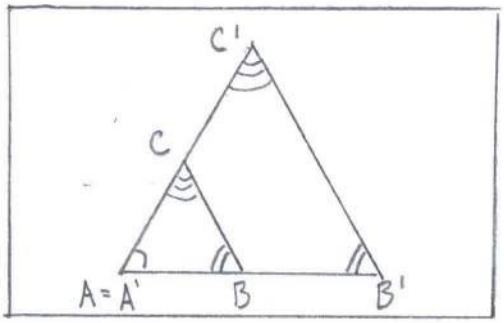
$T(\overline{EF}) = k\overline{E'F'}$ | Proposition 9.1

D) $\triangle A'B'C' \cong \triangle D'E'F'$ | RAA result.

5. (from exercise) "Reflection is a motion"

Proof: Hypothesis	Evidence
1) T is a motion	RAA hypothesis
2) R_m is a reflection across line m	(pg 411) "Definition-reflection"
3) $T(m) = km'$	Proposition 9.1
4) $R_m R_m = R_{m'} R_{m'} = \underline{I}$	Proposition 9.1 (pg 411) "Definition-automorphism"

6.



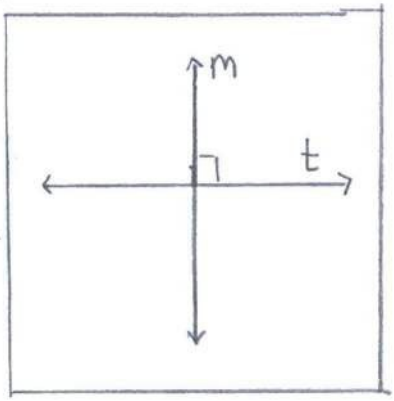
"similarity"

(from exercise) "In the Euclidean plane, $\Delta ABC \sim \Delta A'B'C'$ if and only if there is a similarity sending A, B, C, respectively, onto $A', B', C' \ldots$"

Proof: Hypothesis	Evidence
1) Triangle ΔABC	(pg 43) "Definition-Triangle"
2) The three points A, B, and C are unique to an automorphism $T^{-1} \cdot T = \underline{I}$, $T = T^{-1}$	Lemma 9.2.

3) $T(\triangle ABC) = \triangle A'B'C'$	Corollary 1 to Proposition 9.2
4) $\measuredangle A \cong \measuredangle A'$, $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'}$	Proposition 9.5
5) $\triangle ABC \sim \triangle A'B'C'$	(pg 233) "Chapter 5- Exercise 12"

7.



(from exercise) "The invariant lines of a reflection R_m are m and all lines perpendicular to m."

Proof: Hypothesis	Evidence
1) Two lines $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$ where $\overleftrightarrow{l} \perp \overleftrightarrow{m}$	(pg 24) "Definition- Perpendicular"
2) Transformation T is a reflection across line $\overleftrightarrow{m}$.	Proposition 9.4
3) $T(\overleftrightarrow{m}) = \overleftrightarrow{m'}$	Proposition 9.1
4) Line $\overleftrightarrow{m}$ is invariant under T because $\overleftrightarrow{m} = \overleftrightarrow{m'}$.	(pg 414) "Definition- Invariant"
5) $T(\overleftrightarrow{l}) = \overleftrightarrow{l'}$	Proposition 9.1

6) Line $\overleftrightarrow{l}$ is invariant under T because $\overleftrightarrow{l}=\overleftrightarrow{l'}$ by automorphism.	(pg 414) "Definition- Invariant"

(from exercise) "$R_m \cdot R_K \cdot R_m = R_{K^*}$ where K^* is the reflection of K across $\overleftrightarrow{m}$."

Proof: Hypothesis	Evidence
1) Two lines $\overleftrightarrow{m}$ and $\overleftrightarrow{K}$ are perpendicular	(pg 24) "Definition- Perpendicular"
2) $R_m \cdot R_k \cdot R_m = R_m \circ R_m \circ R_K$ because reflection is a communatative group	Proposition 9.9
3) $R_m R_m R_K = (R_m)^2 R_k$ $= I \circ R_K$	(pg 411) "Definition- Involution"
4) $I \circ R_k = R_{k^*}$	Proposition 9.1b

8. (pg 414) Corollary to Proposition 9.6

"H_A is an involution, and its invariant lines are the lines through A."

Proof: Hypothesis	Evidence
1) Perpendicular lines $\overleftrightarrow{m}$ and $\overleftrightarrow{l}$	(pg 24) "Definition-Perpendicular"
2) A is the intersection between $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$	Proposition 2.1
3) Transformation T half-turns across both lines: $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$: $T = H_A$.	Proposition 9.1
4) For a point $B \neq A$, A is the midpoint to $\overline{B \circ T(B)} = \overline{BB'}$	(pg 167) "Definition-midpoint"
5) $\overline{B \circ T(B)} = \overline{B \circ T^{-1}(B)} = \overline{BB'}$ by involution.	(pg 411) "Definition-Involution"
6) $\overline{BB'} \cong \overline{B'B} \Rightarrow \overline{BB} \cong \overline{BB'}$	(pg 400) "Definition-Invariant"

9. (from exercise) "If an automorphism is

an involution, then it is either a reflection or a half-turn"

Proof: Hypothesis	Evidence
1) A line $\overleftrightarrow{m}$ and point A	(pg ,) "Definition- line, point"
2) Transformation T is an involution; $T = T^{-1}$	(pg 400) "Definition- Invariant"
3) $A \cdot T(A) = A \cdot T(A)^{-1} = \overline{AA'}$	(pg 400) "Definition- Invariant"
4) $\overline{AA'}$ has a mid- point B.	(pg 167) "Definition- midpoint"
5) T is a reflection.	Corollary to Proposition 9.4
6) T is also a half- turn	(pg 414) "Definition- half-turn"

10. (From exercise) "The product $T'T$ of rotations about distinct points can be any of the three types of direct motion."

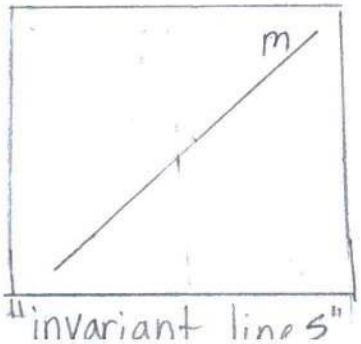
Proof: Hypothesis	Evidence
Reflection:	
1) T is a motion, specially $T_1 = R_\ell R_m$.	(pg 408) "Definition-motion"
2) Line $\overleftrightarrow{\ell}$ passes nearby point A	(pg 12) "Definition-line"
3) Line $\overleftrightarrow{m}$ passes across point A and $\overleftrightarrow{m} \perp \overleftrightarrow{\ell}$.	(pg 12) "Definition-line"
4) $T'T = (R_\ell \circ R_m)' \circ (R_\ell \circ R_m)$ $= \text{Rotation}' \circ \text{Rotation}$	Converse to Proposition 9.8
Translation:	
1) When T is a motion, then T could be $R_\ell R_m$	(pg 408) "Definition-motion"

2) Line $\overleftrightarrow{l}$ is perpendicular to $\overleftrightarrow{m}$ at point A	Proposition 2.1
3) T is also a translation t	Proposition 9.11
4) $T'T = (R_\ell R_m)'(R_\ell \circ R_m)$ $= \text{Rotation}' \circ \text{Rotation}$	Proposition 9.8

Half-turn:

1) T is a motion, $T = R_\ell R_m$	(pg 408) "Definition-motion"
2) Line $\overleftrightarrow{l}$ is perpendicular to line $\overleftrightarrow{m}$ at point A	Proposition 2.1
3) T is a half-turn and rotation	(pg 414) "Definition-half-turn

11.

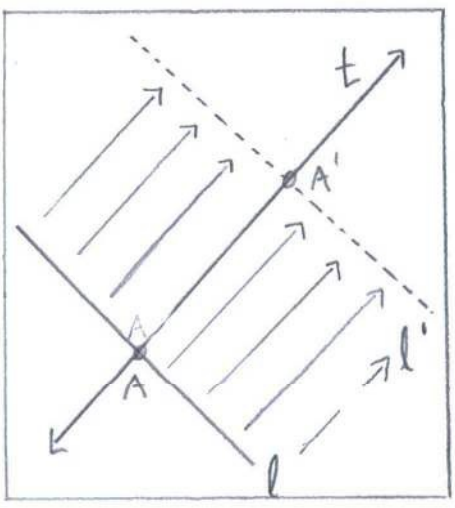


"invariant lines"

(from exercise) "A non-identity rotation that is not a half-turn has no invariant lines."

Proof: Hypothesis	Evidence
1) Line $\overleftrightarrow{l}$ crosses $\overleftrightarrow{m}$ at a point A	Proposition 2.1
2) Motion T is a rotation where $T = R_{\overleftrightarrow{l}} \circ R_{\overleftrightarrow{m}} = H_A$	(pg 408) "Definition-motion"
3) $T(\overleftrightarrow{m}) = \overleftrightarrow{m}'$	RAA false outcome
4) $\overleftrightarrow{m} = \overleftrightarrow{m}'$	RAA false hypothesis ... a contrapositive proof.

12.

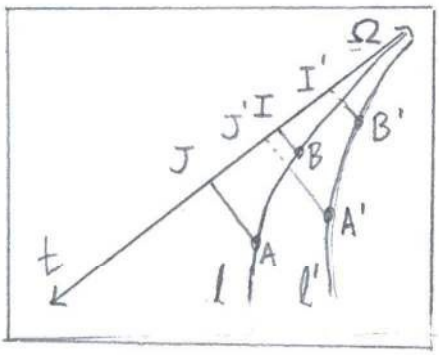


"translation along t"

(from exercise) "$l \parallel l'$"

Proof: Hypothesis	Evidence
1) Line $\overleftrightarrow{l}$ is not invariant to line $\overleftrightarrow{t}$	(pg 414) "Definition-Invariant"
2) A translation across $\overleftrightarrow{t}$ is a motion	Proposition 9.2
3) Point A is the intersection between	Proposition 2.1

$\overleftrightarrow{t}$ and $\overleftrightarrow{l}$.	
4) $T(A) = A'$ $\neq A$ A' lies on $\overleftrightarrow{t}$ and not $\overleftrightarrow{l}$	RAA conjecture
5) $\overleftrightarrow{l}'$ is not invariant	Proposition 9.13
6) $\overleftrightarrow{l} \parallel \overleftrightarrow{l}'$	RAA result



"asymptotically parallel"

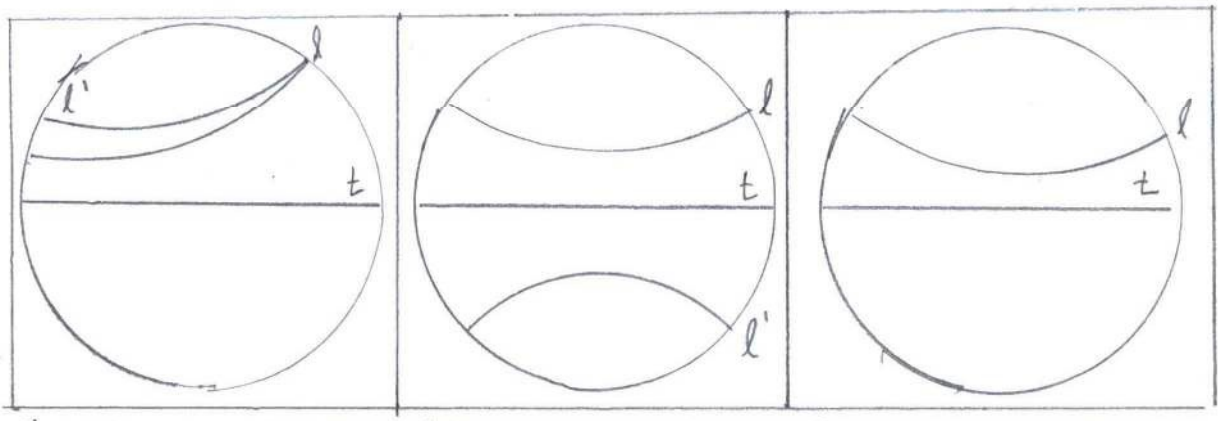
(From exercise) "If $\overleftrightarrow{l}$ is asymptotically parallel to $\overleftrightarrow{t}$ in direction Ω, $\overleftrightarrow{l}'$ is also (in particular, $\overleftrightarrow{l} \parallel \overleftrightarrow{l}'$)"

Proof: Hypothesis	Evidence
1) Line $\overleftrightarrow{l}$ is not invariant to line $\overleftrightarrow{t}$ and $\overleftrightarrow{l}$ is asymptotically parallel to $\overleftrightarrow{t}$	(pg 264) "Definition - Asymptotically Parallel
2) A translation across $\overleftrightarrow{t}$ is a motion T	Proposition 9.2
3) Point A is any point on $\overleftrightarrow{l}$	(pg 12) "Definition - Point"

4) A' lies on $\overleftrightarrow{l'}$ after $T(\overleftrightarrow{l}) = \overleftrightarrow{l'}$ and $T(A) = A'$	RAA hypothesis #
5) B' lies on $\overleftrightarrow{l'}$ from a point B after $T(B) = B'$	RAA hypothesis #2
6) Perpendiculars A, B, A', and B' intersect $\overleftrightarrow{t}$ at I, J, I', J'.	Theorem 6.3
7) Lambert quadrilaterals are congruent, $\square ABJI = \square A'B'J'I'$	(pg 180) "Definition- Lambert Quadrilaterals"
8) because $AB = A'B'$, and $IJ = I'J'$	
9) $\overleftrightarrow{l} \parallel \overleftrightarrow{l'}$ and equidistant from $\overleftrightarrow{t}$ by $AI = A'I'$ and $BJ = B'J'$.	RAA conclusion

(From exercise) "IF $\overleftrightarrow{l}$ is divergently parallel to t, show by diagrams from the

Poincare model that $\overleftrightarrow{l'}$ could meet $\overleftrightarrow{l}$, be divergently parallel to $\overleftrightarrow{l}$, or be asymptotically parallel to $\overleftrightarrow{l}$"



"Intersection" "Divergently Parallel (pg 264)" "Asymptotically Parallel (pg 264)"

13. (pg 423) "Corollary to Proposition 9.15"
"In a Euclidean plane, the product of two translations along different lines is again a translation, and the set of all translations along all lines is a commutative group."

Proof: Hypothesis	Evidence
Translation along different lines:	

1) Unique lines $\overleftrightarrow{m}$ and $\overleftrightarrow{n}$ appear in a Euclidean plane	(pg 12) "Definition-lines"
2) Translations across m and $\overleftrightarrow{n}$ produce motions: T_n and T_m	Proposition 9.2
3) Unique point A is on $\overleftrightarrow{m}$.	(pg 12) "Definition-Points"
4) $T_m(A) = A'$ and $T_n(A') = A''$.	(pg 408) "Definition-motion"
5) $\overleftrightarrow{AA''}$ is a line	(pg 12) "Definition-lines"
6) The third translation is possible along $\overleftrightarrow{AA''}$, $T_{\overline{AA''}}$	Proposition 9.2
7) $T_{\overline{AA''}}(A'') = A$ and an identity to the initial point	Corollary to Proposition 9.10

Commutation:

7) $T_m T_n T_{AA''} = T_n T_m T_{AA''}$ $= I$	Proposition 9.12

14. (From exercise) "In the hyperbolic plane... every direct motion can be expressed as the product of three half-turns"

"In the hyperbolic plane, the product of translations along distinct lines could be any of the three types of direct motions and that two such translations do not commute unless at least one of them is an identity."

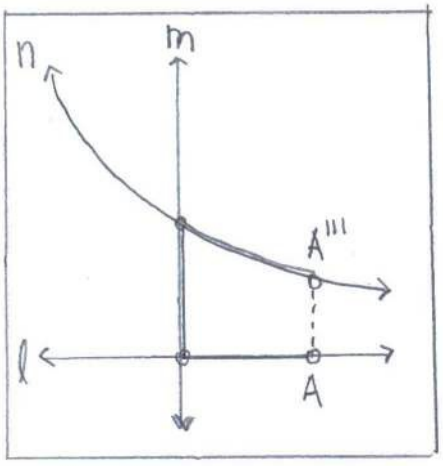
Proof	Hypothesis	Evidence
1)	Lines $\vec{l}, \vec{m}, \vec{n}$ are unique	(pg 12) "Definition - line

Where $\overleftrightarrow{n}$ asymptotes to $\overleftrightarrow{l}$ in a hyperbolic plane.

2) Translations across l, m, n define translations: T_l, T_m, and T_n. — Proposition 9.2

3) Point A resides on $\overleftrightarrow{l}$ — (pg 12) "Definition-point"

Translation:



"Product of Translations"

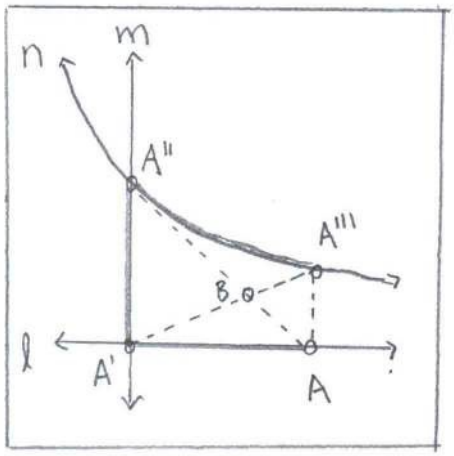
A) $T_n(T_m(T_l(A))) = A'''$ — (pg 408) "Definition-Motion"

B) $\overleftrightarrow{AA'''}$ is another line — Axiom I-1

C) $T_{AA'''}(A''') = A$ and translation is a probable product! — Corollary to Proposition 9.10

Rotation:

A) $T_n(T_m(T_l(A))) = A$ — (pg 408) "Definition-Motion"



"Product of three translations"

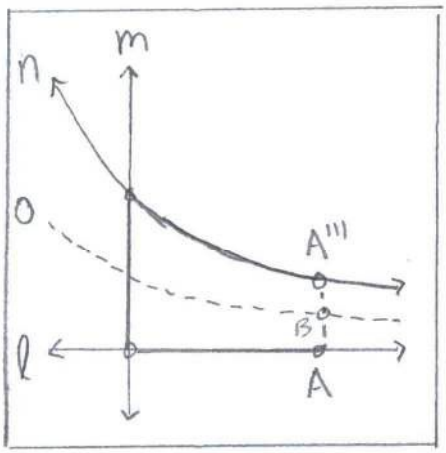
B) $\overline{AA''}$ intersects $\overline{A'A'''}$ at point B with complementary angles $\measuredangle ABA'' + \measuredangle A''BA''' = 180°$ and $\measuredangle ABA' + \measuredangle A'BA'' = 180°$.

Axiom I-2, Proposition 2.1, (pg) "Definition-complementary angles"

c) Rotation about B produces a motion T where $T(A) = A'''$ similar to the product of translations

Proposition 9.8

Reflection:



"Product of three translations"

A) $T_n(T_m(T_\ell(A))) = A'''$

(pg 408) "Definition-Motion"

B) $\overline{AA'''}$ has a midpoint B

(pg 167) "Definition-midpoint"

c) An asymptotically parallel line passes through midpoint B toward $\overleftrightarrow{\ell}$; o.

(pg 264) "Definition-Asymptotically Parallel"

D) A reflection across O, specifically R_O fixes A to A".

Proposition 9.4.

Commutation:

A) $T_\ell \circ T_m \circ T_n \neq T_m \circ T_\ell \circ T_n$

Warning to Proposition 9.9

B) $(T_\ell \circ T_m \circ T_n) \circ (T_\ell \circ T_m \circ T_n)^{-1}$

$= I$, an identity

Proposition 9.1(a)

Identity:

A) $(T_\ell \circ T_m \circ T_n) \circ (T_\ell T_m \circ T_n)^{-1}$

Proposition 9.1(a)

$= T_\ell \circ T_m (T_n)^2 \circ T_m \circ T_\ell$

$= T_\ell \circ T_m \circ I \circ T_m \circ T_\ell$

$= T_\ell (T_m)^2 T_\ell$

$= T_\ell \circ I \circ T_\ell$

$= (T_\ell)^2$

$= I$

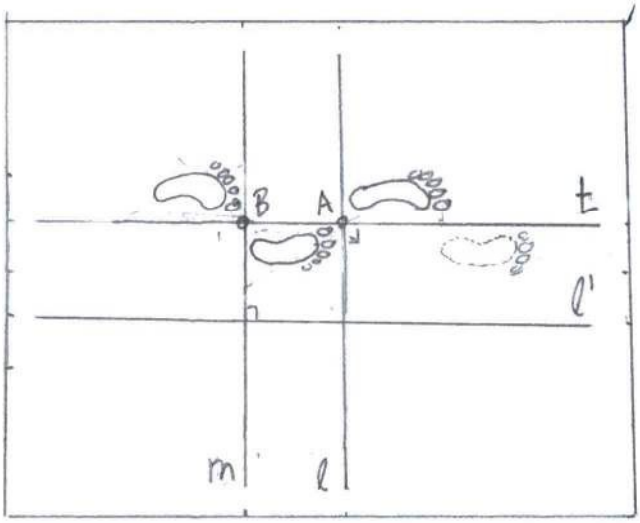
(from exercise) $T=(R_1R_2R_3)^2$ is a translation"

Proof: Hypothesis	Evidence
Identity:	
A) Motion $T=R_1R_2R_3$	Proposition 9.2
B) $T\cdot T^{-1}$ is an involution	(pg 411) "Definition-Involution"
C) $(T^{-1})^2\circ T^2 = \circ\circ\circ$	Proposition 9.1a
$\circ\circ\circ = ((R_1R_2R_3))^{-1}{}^2\circ(R_1R_2R_3)^2$	
$=(R_3R_2R_1)^2\circ(R_1R_2R_3)^2$	
$=R_3^2R_2^2(I)^4R_2^2R_3^2$	
$=R_3^2(I)^8R_3^2$	
$=(I)^{12}$	
$=I$	
D) $T=(R_1R_2R_3)^2$	Corollary to Proposition 9.5
Product of Reflections (with Identity):	

A) $\Delta ABC \cong \Delta A'B'C'$	RAA conjecture
B) Point $A \neq A'$	(pg 12) "Definition-Point"
C) Line 1 is a perpendicular bisector to $\overline{AA'}$	Proposition 4.5
D) A reflection (R_1) reflects A onto A', B onto B', and C to C'.	(pg 411) "Definition-reflection"
E) Point $B \neq B'$, so $AB \cong \overline{A'B'}$	Axiom C-1
F) Line 2 is a perpendicular bisector to $\overline{BB'}$	Proposition 4.5
G) Reflection $R_1 R_2$ sends $\overline{AB}$ to $\overline{A'B'}$	(p 411) "Definition-reflection"
H) Point $C \neq C'$, so $\overline{AC} \cong \overline{A'C'}$ and $\overline{BC} \cong \overline{B'C'}$	Axiom C-1

I) Line 3 is across $A'B'$	(pg 411) "Definition-reflection
J) $R_1 R_2 R_3$	Corollary 1 to Proposition 9.2
K) $T = (R_1 R_2 R_3)^2$ is also a translation	RAA result

16.



"Glides"

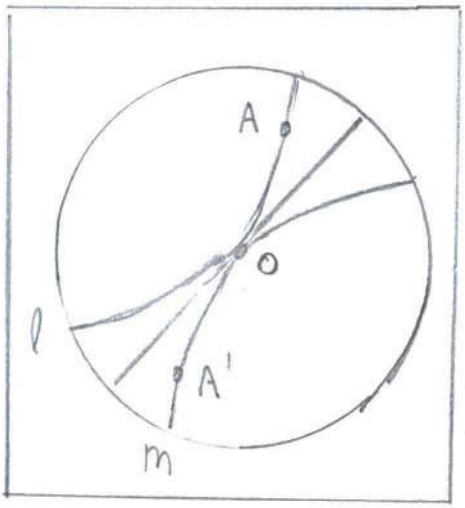
(from exercise) "Let T, T' be glides along perpendicular lines,... $(T \circ T^{-1})$ is a half-turn if and only if the plane is Euclidean"

Proof: Hypothesis	Evidence
1) Lines be perpendicular: $\overleftrightarrow{\ell} \perp \overleftrightarrow{\ell'}$ and $\overleftrightarrow{t} \perp \overleftrightarrow{m}$.	(pg 24) "Definition-Perpendicular"
2) $T = R_\ell R_m$, $T' = R_t T$	Proposition 9.18
3) $(T' \circ T^{-1}) = R_t (T \circ T^{-1}) R_t$	(pg 426) "Definition-

$= H_A \circ R_m \circ R_\ell \circ H_B$ | Glide"

$= I$

17. (from exercise) "In the hyperbolic plane... every direct motion can be expressed as a product of three half-turns."



"Reflection"

Proof: Hypothesis	Evidence
1) In a hyperbolic plane, on a Poincare disc, lines $\overleftrightarrow{\ell}$ and $\overleftrightarrow{m}$ describe direct motion.	RAA conjecture
2) $T = H \cdot H \cdot H$	Proposition 9.2
Rotation:	
A) $\overleftrightarrow{\ell}$ and $\overleftrightarrow{m}$ intersect at the center O	Proposition 2.1
B) Point A is on line $\overleftrightarrow{\ell}$	(pg 12) "Definition-Point"

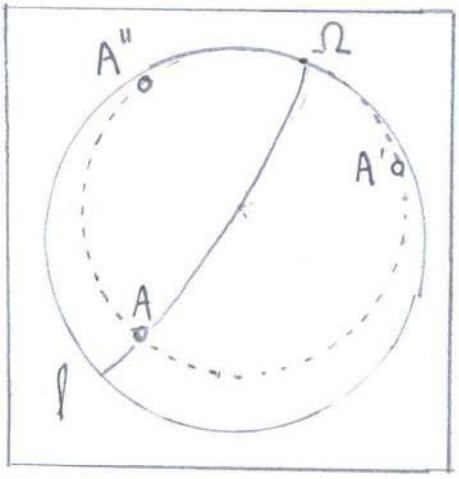
C) $T = H_o H_o H_o$ | RAA result

$= (H_o)^2 H_o$

$= I \circ H_o$

$= R_o$

Parallel Displacement:



"Parallel Displacement"

A) Line $\overleftrightarrow{l}$ meets ideal point Ω | (pg 422) "Definition- Ideal Point"

B) Point A is on line $\overleftrightarrow{l}$ | (pg 12) "Definition- Point"

C) Three half-turns is a parallel displacement and direct motion | RAA result

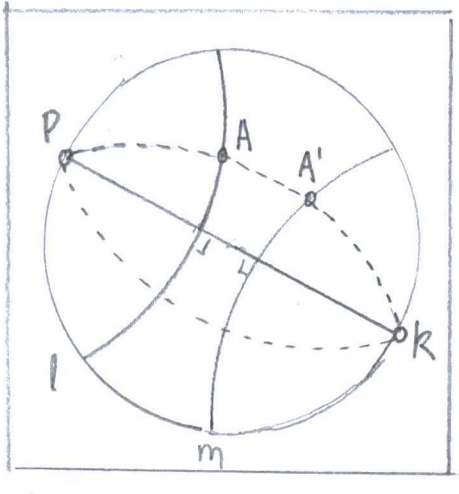
Translation:

A) Lines $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$ are parallel | (pg 304) "Definition- Poincare Parallel"

B) A $\overleftrightarrow{k}$ common perpendicular resides between | (pg 305) "Definition-

both $\overleftrightarrow{l}$ and $\overleftrightarrow{k}$.

c) Point A is on line $\overleftrightarrow{l}$



"Translation"

D) A translation is three half-turns about point P.

Poincare Perpendicular"

(pg 12) "Definition-Point"

RAA result

10. (From exercise) "If T is a glide and $n \neq 0$ ... $T^n \neq I^n$."

Proof: Hypothesis

1) Motion is a glide where $T = (\text{reflection})(\text{translation}) = R_t \cdot T$

Evidence: (pg 426) "Definition-glide"

Note: Transformations and Translations have a letter T.

Base case:

A) $n = 1$

B) $T^1 = (R_t \circ T)^1 \neq I$

RAA guess
RAA result

First Step:

A) $n = 2$

B) $T^2 = (R_t \circ T)^2 = R_t^2 \circ T^2$
$= I \circ T^2$
$\neq I$

RAA guess
RAA result

Inductive Step:

A) $n = k+1$

B) $T^{k+1} = (R_t \circ T)^{k+1}$
$= R_t^{k+1} \circ T^{k+1}$
$= R_t^{2m} \circ R_t \circ T^{2m} \circ T$
$\neq I$

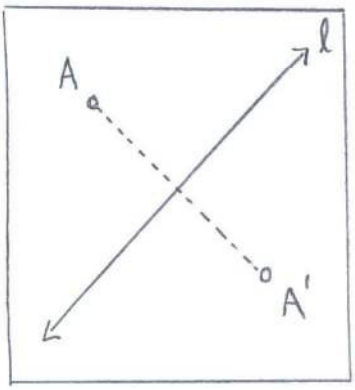
RAA guess
RAA result

Solutions don't exist for even or odd numbers.

19. (from exercise) "IF rotations T, T^{-1} about O have the same effect on point $P \neq O$ ... $T = T'$"

Proof: Hypothesis	Evidence
1) $T = T'$	RAA conclusion
2) O and P are unique points	(pg 12) "Definition-point"
3) $T = R_\circ$	Proposition 9.2
4) $T(P) = P'$	(pg 408) "Definition-Motio
5) $T^{-1}(P) = P'$	(pg 408) "Definition-Motion"
6) $T(P) = T^{-1}(P)$ effected P equally	RAA hypothesis by converse argument.

20. (from exercise) "IF T is any motion, A any point, and $\overleftrightarrow{l}$ any line ... $T \cdot H_A \cdot T^{-1}$ is a half-turn and that $T \cdot R_\ell \cdot T^{-1}$ is a reflection."



Proof: Hypothesis	Evidence
1) By information given	RAA conjecture

2) $\overline{T}$ is commutative

3) $T \circ H_A \circ T^{-1} = H_A \circ T \circ T^{-1}$

$= H_A(I)$

$= H_A$

4) $T \circ R_\ell \circ T^{-1} = R_\ell \; T \circ T^{-1}$

$= R_\ell(I)$

$= R_\ell$

5) $T \circ H_A \circ T^{-1}$ is a half-turn and $T \circ R_\ell \circ T^{-1}$ is a reflection.

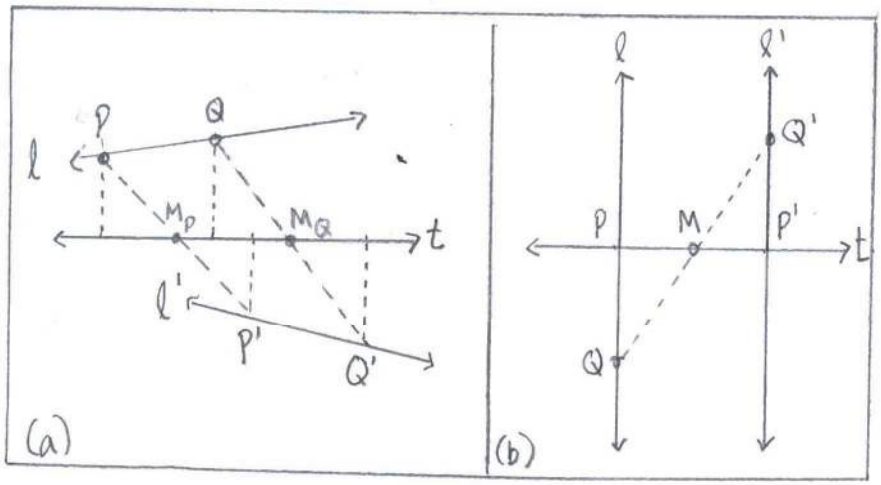
(pg 400) "Definition - commutative"

(pg 399) "Definition - Inverse"

(pg 399) "Definition - Inverse"

RAA result

21.



"Hjelmslev's lemma"

(pg 459) "Hjelmslev's Theorem"

"Let $\overleftrightarrow{\ell}$ and $\overleftrightarrow{\ell'}$ be distinct lines and let T be a motion transforming $\overleftrightarrow{\ell}$ onto $\overleftrightarrow{\ell'}$. As Point P varies on $\overleftrightarrow{\ell}$ and its image P' under T varies on $\overleftrightarrow{\ell'}$, the midpoints of the

Segments PP' either are distinct and collinear or else they all coincide."

Proof: Hypothesis	Evidence
1) From the idea above, a hypothesis forms.	RAA conjecture
2) Q is a point on $\overleftrightarrow{l}$ and Q' another point on $\overleftrightarrow{l'}$	(pg 12) "Definition-point"
Case 1: Figure a	
A) $\overleftrightarrow{m}$ and $\overleftrightarrow{n}$ are the feet to the perpendicular from P to $\overleftrightarrow{E}$ and Q to $\overleftrightarrow{E}$, respectively.	Proposition 4.4b
B) m' and n' are the feet opposite side the midpoint from P' to $\overleftrightarrow{E}$ and Q' to $\overleftrightarrow{E}$, respectively	(pg 110) "Definition-opposite side"

C) $\measuredangle PmM_P \cong \measuredangle P'm'M_P$ and $\measuredangle QnM_Q \cong \measuredangle Q'n'M_Q$	Proposition 3.15
D) $P*M_P*P'$ and $Q*M_Q*Q'$ are collinear	(pg 70) "Definition-collinear"
E) $PM_P \cong P'M_P$ and $QM_Q \cong Q'M_Q$	RAA result
F) M_P and M_Q are midpoints on axis $\overleftrightarrow{t}$	(pg 167) "Definition-midpoints"
Case 2: Figure b.	
A) P and P' are opposite midpoint M, along with Q and Q' opposite midpoint M.	(pg 110) "Definition-opposite sides"
B) PP' crosses QQ' at point M	Proposition 2.1
C) M is the midpoint on PP' and QQ'	RAA result

22. (From exercise) "Hjelmslev's Theorem ... by glide"

Proof: Hypothesis	Evidence
1) By information provided.	RAA hypothesis
2) Q is on $\overleftrightarrow{l}$ and Q' on l'	(pg 12) "Definition-point"
3) $T = H_M \cdot R_l$ is a glide	Proposition 9.18
4) T is an opposite motion	(pg 429) "Definition-opposite motion"
5) Midpoint PP' and QQ' are on $\overleftrightarrow{t}$	Hjelmslev's theorem
6) Midpoints are collinear or else coincide.	RAA result.

23. "Orientation" aids in specific contexts, including the number line, direction, and motion. Direct motion requires no "orientation": translation, rotation, and parallel displacement. While, opposite motion needs "orientation." A motion, such as reflection or glide produce relative position, especially to polygon with symbols on vertices.

24. (pg 436) Proposition 9.24

"In the Cartesian model of the Euclidean plane, a similarity is represented in complex coordinates either in the form:

$$z \longrightarrow w_0 z + z_0 \qquad w_0 \neq 0$$

(in which case it is called direct) or in the form:

$$z \longrightarrow w_0 \bar{z} + z_0 \qquad w_0 \neq 0$$

(in which case it is called opposite). The direct similarities form a four-parameter group."

Proof: Hypothesis	Evidence
Direct Similarities:	
Translation:	
$z + c \to w_0 z + z_0 + c \to w_0 z + (z + c)_0$	Lemma 9.1, $z + c$ implies automorphism
Rotation:	
$e^{w_0 \theta} \lvert z \rvert \to (x + w_0 y) \lvert (w_0 z + z_0) \rvert$ $\to (x+y)z + (x-y)z_0$	Lemma 9.1, $e^{w_0 z}$ implies automorphism

Dilation

$cz \to c_0(w_0 z + z_0)$

$\to (wc)_0 z + (zc)$

Lemma 9.1 shows an automorphism.

Opposite motion:

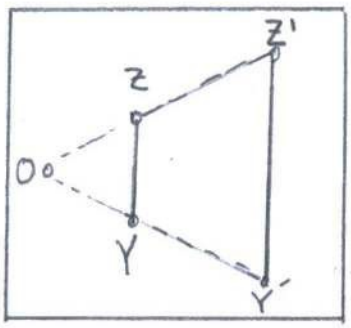
Reflection:

$z \to -(w_0 z + z_0)$

$\to -w_0 z - z$

(pg 58) "Definition- Reductio Ad Absurdia"

25.



"Dilation"

(from exercise) "Every similarity that is not a motion has a unique fixed point."

Proof: Hypothesis	Evidence
1) Dilation is not a direct motion - translation, rotation, or parallel displacement	(pg 429) "Definition- Direct motion"
2) But dilation is a direct similarity	Proposition 9.24

3) A dilation scales points on segment YZ. $Z' = C(Z) \to C(w_0 Z + Z_0) \to (w_0 C)Z + CZ_0$ $Y' = C(Y) \to C(w_0 y + y_0) \to (cw_0)y + cy_0$	(pg 321) "Definition-Dilation"
4) $O * Z * Z'$, $O * Y * Y'$	(pg 70) "Definition-Collinear
5) $\overline{OZ'}$ intersects $\overline{OY'}$ at point O	Proposition 2.1

26. (from exercise) "Reflections in the Cartesian model are characterized among all opposite motions $Z \to e^{i\theta}\bar{Z} + Z_0$ by the equation $e^{i\theta}\bar{Z}_0 + Z_0 = 0$ and that the axis of such a reflection is the line $Z(yy_0 + XX_0) = X_0^2 + y_0^2$ if $Z = X_0 + iy_0 \neq 0$, whereas $Z_0 = 0$, it is a line

$y\cos(\theta/2) - x\sin(\theta/2) = 0$

Proof: Hypothesis	Evidence
Axis of reflection perpendicular ($m = \frac{-x_0}{y_0}$ with midline $(\frac{x_0}{2}, \frac{y_0}{2})$):	Point-Slope Formula: $y - y_1 = m(x - x_1)$
1) $y - \frac{y_0}{2} = -\frac{x_0}{y_0}(x - \frac{x_0}{2})$	(pg 411) "Definition-Fixed Points"
becomes $2(yy_0 + xx_0) = x^2 + y^2$	
Axis of reflection parallel ($m = \frac{y_0}{x_0} = \frac{\sin(\theta/2)}{\cos(\theta/2)}$ with midline $(0,0)$):	
1) $y - 0 = \frac{\sin(\theta/2)}{\cos(\theta/2)}(x - 0)$	(pg 411) "Definition-Fixed Points"
becomes $y\cos(\theta/2) - x\sin(\theta/2) = 0$	

27. Dilation is a direct motion with the form: $kZ \rightarrow R(w_0 Z + Z_0)$. Z is a complex-coordinate, in addition to Z_0. Dilation ratio k scales complex coordinates. A final representation could be $k(e^{i\theta} Z + Z_0 - Z_1)$ for center Z_1 after dilation.

28. (from exercise) "A coordinate-free description is given as follows: A similarity is direct (respectively, opposite) if and only if it is the product of a dilation with a direct (respectively, opposite) motion."

Proof without coordinates:

Similarities (pg 436)	Motion
Direct	Dilation • Rotation • Translation
Opposite	Reflection × Direct Similarity

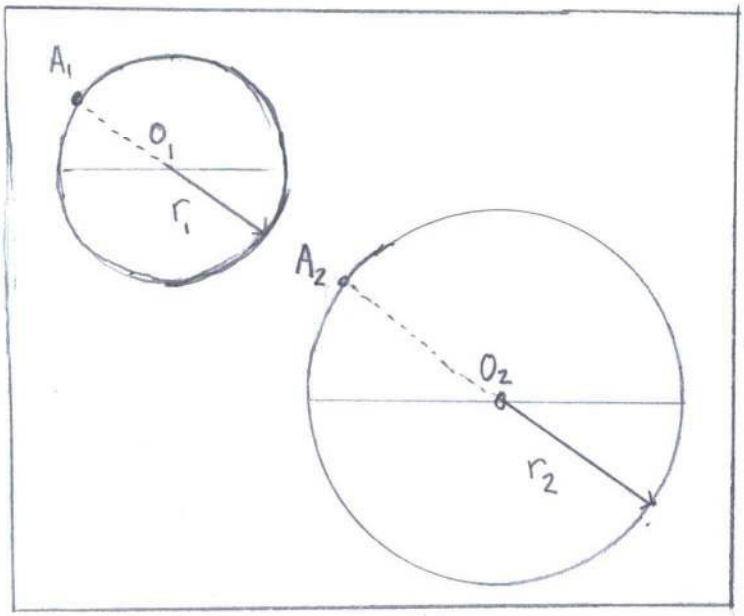
(from exercise) "If AB and A'B' are any two segments, there is a unique direct similarity taking A to A', and B to B'."

Proof:

Hypothesis	Evidence
1) Segment AB	(pg 16) "Definition-segment"

2) Dilations are a direct similarity. $R(AB) \rightarrow A'B'$	(pg 430) "Definition-Direct Similarity"

29.

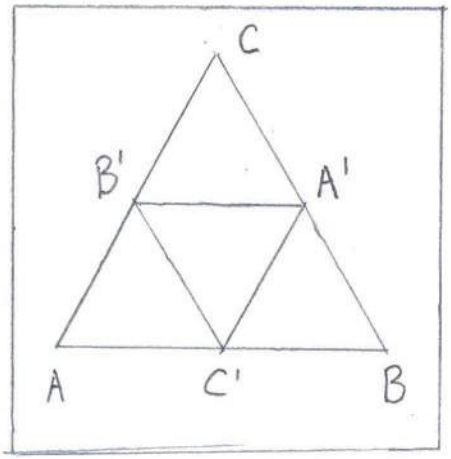


(from exercise) "There are two distinct dilations D_1, D_2 which transform γ_1 to γ_2"

Proof:

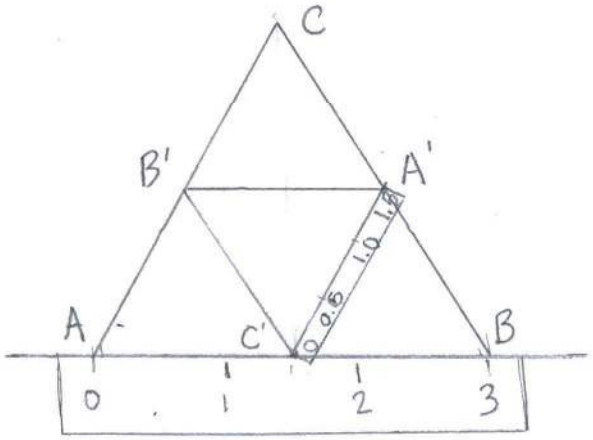
Hypothesis	Evidence
1) Circle γ_1 has center O_1 with radius r_1, and γ_2 with center O_2 and radius r_2. Lastly, O_1 is not O_2!	(pg 17) "Definition-circle"
2) Point A_1 on γ_1 forms $\overline{O_1A_1}$ parallel to $\overline{O_2A_2}$	Euclid's Postulate V
3) $\overline{O_2A_1}$ is an image of $\overline{O_1A_1}$ by D_1	(pg 436) "Definition-Dilation"
4) $\overline{O_2A_2}$ is an image of $\overline{O_1A_1}$ by D_1	(pg 436) "Definition-Dilation"

30.



"Unique dilation ratio"

(From exercise) "Show there is a unique dilation of ratio -1/2 taking ΔABC onto $\Delta A'B'C'$."



31. (from exercise) "Show that the set of all dilations (with all the possible centers and ratios) is not a group, whereas the set of all translations and dilations (of a Euclidean plane) is a group and that group is non-commutative."

(pg 431) Translation:

$$T(x,y) = (x+e, y+f)$$

Dilation:

$$D(x,y) = (R(x-c)+c,\ y)$$
$$= (Rx+(1-R)c,\ y)$$

Dilation fails composite properties necessary for sets:

$$D_2(D_1(x,y)) = ((k_2(k_1x+(1-k_1)c_1)+(1-k_2)c_2,\ y))$$
$$= (k_1k_2x + k_2(1-k_1)c_1 + (1-k_2)c_2,\ y)$$

If $k_1k_2 = 1$ and $c_1 \neq c_2$, where $k_1 \neq 1$, then

$$D_2(D_1(x,y)) = (\underbrace{x + (1-k_1)(c_1-c_2)}\ ,\ y)$$

"Translations form, not Dilation!"

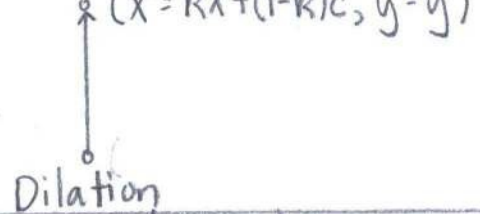
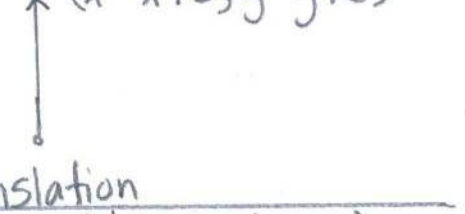
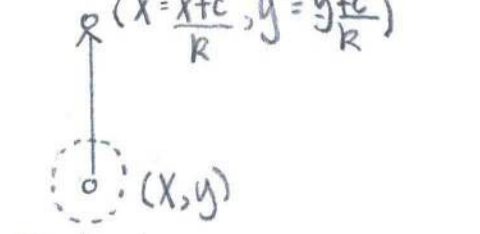
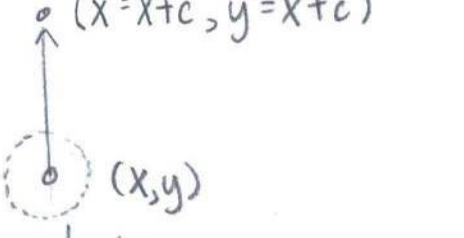
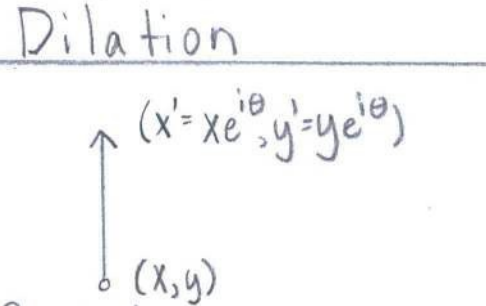
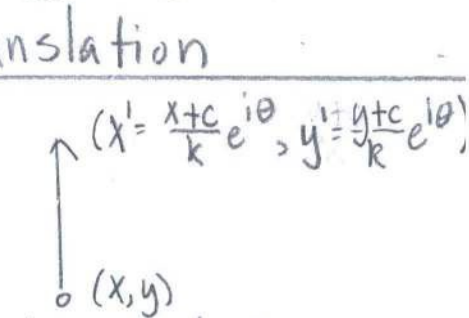
Also, $D(T(x,y))$ and $T(D(x,y))$ become an automorphism, while $D(T(x,y)) \neq T(D(x,y))$, so not commutative by page 400.

32. (from exercise) "Dilations are geometrically characterized among all similarities by the two properties of (i) mapping each line $\overleftrightarrow{l}$ onto a line equal to or parallel to $\overleftrightarrow{l}$; and (ii) having a fixed point."

Proof: Hypothesis	Evidence
1) Dilation in X-coordinates: $D(x,y)=(Rx+(1-R)c, y)$	RAA conjecture
Case 1: a line $y=mx+b$:	
A) $D(x,y)=D(x,mx+b)$ is a parallel line after dilation. But $D(x,y)$ shifts by an independent variable when $m \neq 0$.	RAA result.
Case 2: Point fixed at $(0,0)$:	

A) $D(x_y) = D(0,0)$ is a point when $c=0$ | RAA result

33. (from exercise) "Show an automorphism which (i) fixes every point at infinity is either a dilation or a translation; (ii) fixes two points at infinity is an opposite similarity (if ℓ and m determine the two fixed points, then $\ell \perp m$); (iii) has no fixed points at infinity is either a rotation that is not an identity and not a half-turn, or the product of a rotation with a dilation."

(i)		
(ii)		
(iii)		

34. (From exercise) "An affine transformation is a collineation."

Proof: Hypothesis	Evidence
1) $Z \longrightarrow AZ + Z_0$ is an affine transformation where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	(pg 462) "Definition- Affine Transformation"
2) A has multiple permutations and by definition (pg 398) collineation to Z.	(pg 398) "Definition- collineation"

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$= \begin{bmatrix} a_{22} & a_{21} \\ a_{12} & a_{11} \end{bmatrix}$$

$$= \begin{bmatrix} a_{22} & a_{12} \\ a_{21} & a_{11} \end{bmatrix}$$

(From exercise) "The set of all affine transformations is a group (called

the affine group].

Proof: Hypothesis	Evidence
1) $Z \rightarrow AZ + Z_0$ is an affine transformation where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	(pg 462) "Definition- Affine Transformati
Requirement #1: Composition	
A) $1, Z \in G \Rightarrow 1(z) \in G$	(pg 399) "Composite Transformation
Requirement #2: Identity	
A) $I \in G$ where $I \circ Z = Z = zI$ and $z \in G$	(pg 399) "Definition- Identity"
Requirement #3: Inverse	
A) $Z \in G \Rightarrow Z^{-1} \in G$ where $Z \cdot Z^{-1} = I = Z^{-1} \cdot Z$	(pg 399) "Definition- Inverse"
Requirement #4: Association	

A) $1(z \circ z^{-1}) = (1 \circ z) z^{-1}$ for all $1, z, z^{-1}$ (pg 399) "Definition-Associative Law"

35. (pg 86) "Algebraic models of affine and Projective Planes."

F = field

F^2 = set of all pairs (x, y) on F

= affine plane

Affine Plane:

$F^2 = [(1,1), (0,1), (2,2), (1,2), (0,0), (-1,0), \ldots]$

Solutions to parallel lines: $x - y + 1 = 0$

$x - y = 0$

Collineation of an Affine plane (F^2) by Affine Transformation (T):

$T^* = T * F^2$ is an argument about equivalent permutations (pg 398) in a set, specifically F^2. The coordinate pairs are still solutions to parallel lines; $x - y + 1 = 0$ and $x - y = 0$.

36. (From exercise) "Möbius Transformation"

$$T: z \longrightarrow \frac{az+b}{cz+d} \quad \text{where } \delta = ad-bc \neq 0, \text{ "Non-zero Determinant"}$$

(From exercise) "A transformation with $c \neq 0$ can be factored into a composite."

$$T = T_1 \circ T_2 \circ T_3 \circ T_4 \quad \text{where } T_1 = \text{"Translation"} = z \to z + (a/c)$$

$$= T_1(T_2(T_3(T_4)))$$

$$= T_1(T_2(T_3(z+(d/c))))$$

$$= T_1\left(T_2\left(\frac{1}{z+d/c}\right)\right)$$

$$= T_1\left(\frac{-\delta/c^2}{z+d/c}\right)$$

$$= \frac{a}{c} - \frac{\delta/c^2}{z+d/c}$$

T_2 = "Similarity" $= z \longrightarrow (-\delta/c^2)z$

T_3 = "Mapping" $= z \to z^{-1}$

T_4 = "Translation" $= z \longrightarrow z + (a/c)$

$$T_4 T_3 T_2 T_1 = \left(\frac{a[z+(a/c)+b]}{c[z+(a/c)]+d}\right)\left(\frac{a\left[\frac{-\delta}{c^2}\right]z+b}{c\left(\frac{-\delta}{c^2}\right)z+d}\right)\left(\frac{az^{-1}+b}{cz^{-1}+d}\right)\left(\frac{c(z+d/c)+b}{c(z+d/c)+d}\right)$$

Projective Transformation ($P^2(F)$):

$T^{x'} = B \cdot P^2(F)$ where $B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and $P^2(F) = [x\ y\ z]$ is a projective transformation for the solutions to $ax + bx + cz = 0$. The coefficient B solves lines $x - y + 1 = 0$ and $x - y = 0$. The equation preserves incidence. Incidence axiom #1 requires two points to every line; (1,1,0) and (1,0,0). A line $x - y = 0$ defines two points; Incidence Axiom #2. Lastly, in Incidence Axiom #3, no three points form a line; (0,0,0), (1,1,0), (0,1,0); because two parallel lines.

37. (from exercise) "A Möbius Transformation maps the set of all circles and lines in a Cartesian model onto itself, preserving orthogonality."

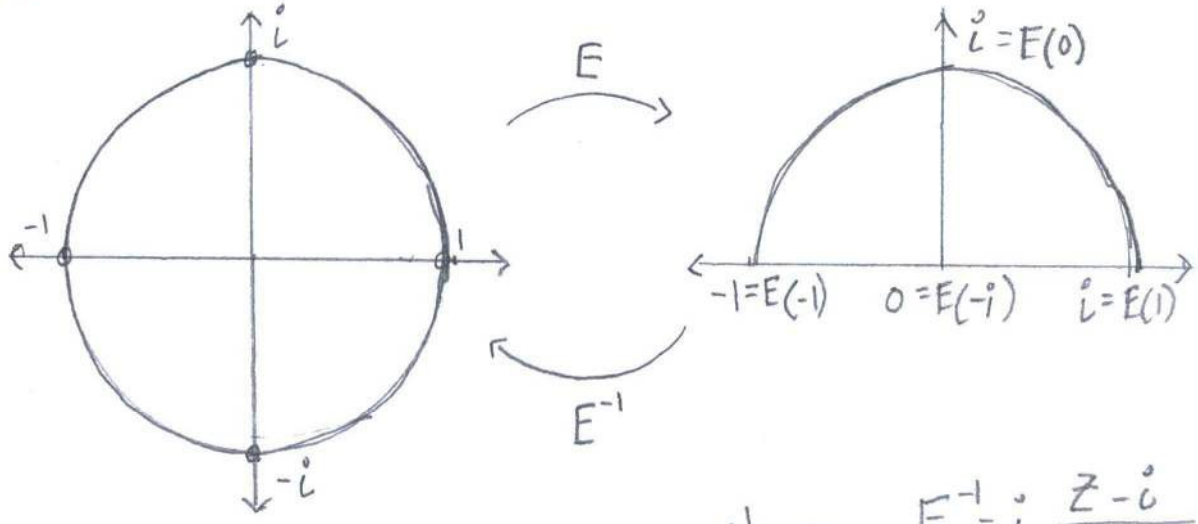
$$T = T_4 \circ T_3 \circ T_2 \circ T_1$$

$$= \left(\frac{a[z+(a/c)]+b}{c[z+(a/c)]+d}\right)\left(\frac{a\left[\frac{-\delta}{c^2}\right]z+b}{c\left[\frac{-\delta}{c^2}\right]z+d}\right)\left(\frac{az^{-1}+b}{cz^{-1}+d}\right)\left(\frac{c(z+d/c)+b}{c(z+d/c)+d}\right)$$

By composition, T maps circles onto circles through inversion.

Proof: Hypothesis	Evidence
1) Circle γ has center z	Euclid's Postulate III
2) A point z^{-1} describes the inverse from center z.	Proposition 7.5
3) An 'inverse "circle" γ^{-1} to γ exists from z^{-1}	Proposition 7.9

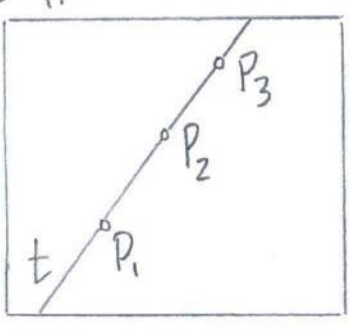
38. (from exercise) "The mapping $E: z \to i\,\frac{i+z}{i-z}$ maps the open unit disk one-to-one onto the upper half-plane, sends i to ∞, and maps all the other points of the unit circle onto the x-axis"



where $E^{-1} = i\,\frac{z-i}{z+i}$

The complex i "sends" points to infinity when the dependent-variable $E(i) = i\,\frac{i+1}{i-1} = \infty$.

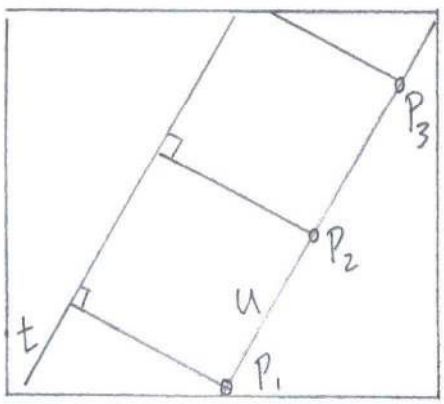
39.



(from exercise) "G be the group of all translations along a line $\overleftrightarrow{t}$. IF P lies on $\overleftrightarrow{t}$, prove $G \cdot P = \overleftrightarrow{t}$"

Proof: Hypothesis	Evidence
1) $T \in G, I \in G, T^{-1} \in G,$ and $(T \cdot I)T^{-1} = T(I \cdot T^{-1})$	(pg 399) "Definition- Group defined by composition, identity, inverse, and association."
2) $T(P) = T^{-1}P$ $= \overleftrightarrow{t}$ or $T(P_1) = T(P_2)$ $= T(P_3)$ $= \circ\circ\circ$ $= T(P_n) \neq I$ $= \overleftrightarrow{t}$	Lemma 9.4

40.



(From exercise) "P doesn't lie on $\overleftrightarrow{t}$. Show that G•P is a unique parallel to $\overleftrightarrow{t}$ through P in the case where the plane is Euclidean."

Proof: Hupothesis	Evidence

1) $T \in G$, $I \in G$, $T^{-1} \in G$, and $(TI)T^{-1} = T(IT^{-1})$ (Pg 399) "Definition-Groups"

2) $T(P) = T^{-1}(P)$ Lemma 9.4

$= \overleftrightarrow{u}$. or

$T(P_1) = T(P_2)$

$= T(P_3)$

$= \cdots$

$= T(P_n) \neq I$

$= \overleftrightarrow{u}$

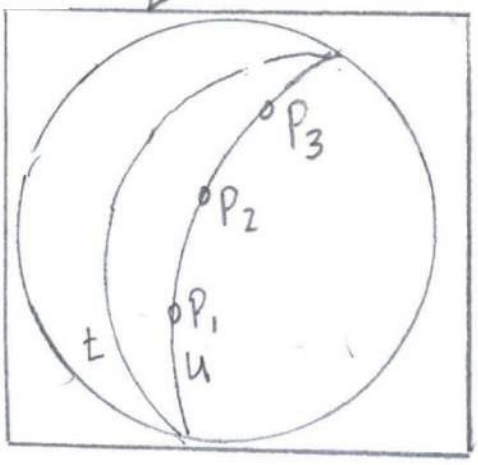
3) Perpendicular bisectors exist from P_1 to $\overleftrightarrow{t}$ and P_n to $\overleftrightarrow{t}$ Proposition 4.4b

4) Perpendicular bisectors have the same length because parallels Measurement Theorem 4.3

(from exercise) "Show $G \cdot P$ is the equidistant curve to $\overleftrightarrow{t}$ through P if the plane is hyperbolic."

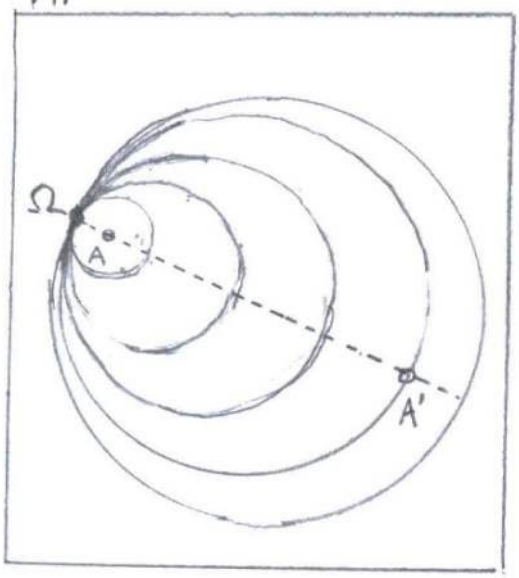
Proof:

"Equidistant Curve"



Hypothesis	Evidence
1) $T \in G, I \in G, T^{-1} \in G,$ and $(TI)T^{-1} = T(IT^{-1})$	(pg 399) "Definition-Group"
2) $T(P) = T^{-1}(P)$ $= \overleftrightarrow{t}$ or $T(P_1) = T(P_2)$ $= T(P_3)$ $= \cdots$ $= T(P_n) \neq I$ $= \overleftrightarrow{t}$	Lemma 9.4
3) Perpendicular bisectors exist from P_1 to $\overleftrightarrow{t}$ and P_n to $\overleftrightarrow{t}$.	Chapter 4.4b

41.

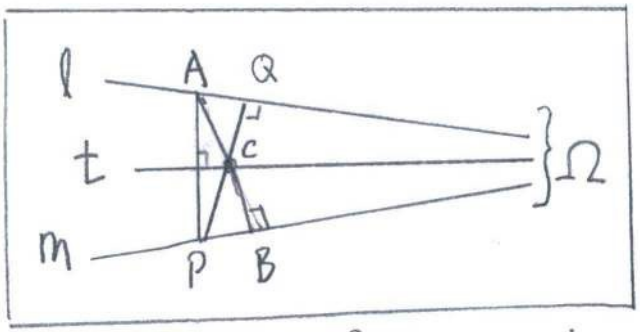


"Horocycle"

(from exercise) "A and all points A' such that the perpendicular bisector of $\overline{AA'}$ has Ω as an end."

Proof: Hypothesis	Evidence
1) In a hyperbolic plane, A is any point.	(pg 12) "Definition- Point"
2) An equidistant circle surrounds point A with radius $\overline{A\Omega}$.	Euclid's Postulate III
3) Multiple parallel limits circle with center A	(pg 257) "Definition- Limiting Parallel"
4) Point A' is on a parallel line $\overleftrightarrow{A\Omega}$	Proposition 2.1
5) AA' forms a perpendicular bisector at Ω	Proposition 9.17b

42.



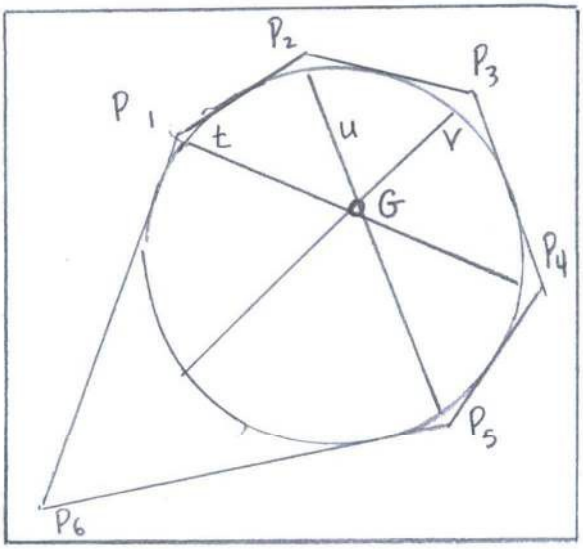
"Corresponding Points and Symmetry axis"

(from exercise) "t is a symmetry axis of l and m."

Proof: Hypothesis	Evidence
1) Line $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$ are asymptotically parallel to Ω	(Pg 264) "Definition-Asymptotically Parallel"
2) A is incident to $\overleftrightarrow{l}$	(pg 12) "Definition-Incident"
3) Point B is foot to the perpendicular from A to m.	Proposition 4.4b
4) A unique point P on $\overleftrightarrow{m}$ such $\overline{PQ} \cong \overline{AB}$ where Q is a foot to a perpendicular to $\overleftrightarrow{l}$	Proposition 4.4b
5) Line $\overleftrightarrow{E}$ is the perpendicular bisector of $\overline{AP}$.	Proposition 4.4b
6) $\triangle APB \cong \triangle PAQ$	Axiom C-6 (S.A.S.)
7) $\square ABPQ$ is a Saccheri Quadrilateral	(pg 177) "Definition-Saccheri Quadrilateral"

8) The line t is a perpendicular bisector to both $\overline{AP}$ and $\overline{QB}$.	Proposition 4.4b
9) $R_t(\ell) = m$ because the reflection fixes points and segments: $A = P, Q = P_1$ and $\overline{AQ} = \overline{PB}$.	Lemma 9.1

43.



(from exercise) "The symmetry axes t, u, v of the pairs of sides of a trebly asymptotic triangle are concurrent in a point G"

"Brianchon's Theorem"

Proof:

Hypothesis	Evidence
1) Lines t, u, v cross an irregular	(pg 32) "Definition- convex polygon"

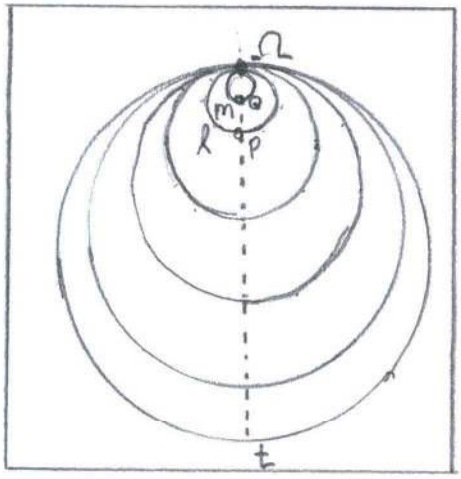
hexagon at vertice pairs (1,4), (2,5), and (3,6).	
2) Lines $\overleftrightarrow{t}$, $\overleftrightarrow{u}$, and $\overleftrightarrow{v}$ are angle bisectors	Proposition 4.4a (Bisectors)
3) Point G is the incenter to the hexagon.	(pg 197) "Definition-Incenter"
4) $R_t(\vec{u}) = \vec{v}$ because $G = R_t(G)$	Lemma 9.1

(from exercise) $R_v \circ R_u \circ R_t = R_u$

Proof: Hypothesis	Evidence
1) A regular hexagon looks dihedral and finite to reflection. But irregular hexagons are difficult interpretation by art as a dihedral. The incenter G is	Lemma 9.6 Lemma 9.9

Concurrent to the hexagon and in a constant circle.

44.

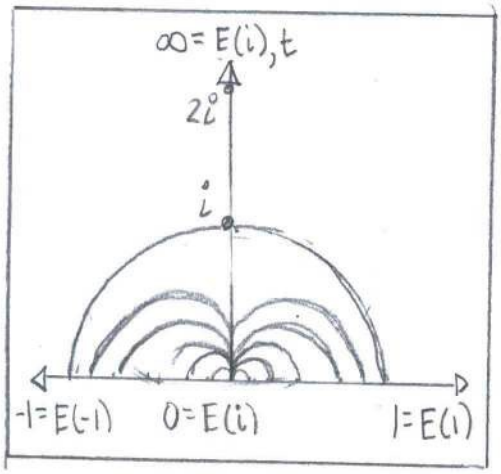


"Horocycles are congruent to one another"

(From exercise) "Horocycles are congruent to one another."

Proof: Hypothesis	Evidence
1) Points P and Q are on line $\overleftrightarrow{t}$	Axiom I-1
2) Ω is an ideal point	(pg 422) "Definition- Ideal Point"
3) Line $\overleftrightarrow{l}$ in a hyperbolic plane has radius $\overline{P\Omega}/2$	Euclid's Postulate III
4) Line m has radius $\overline{Q\Omega}/2$	Euclid's Postulate III
5) Both $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$ are incident to point Ω	Proposition 2.1
6) Points P and Q correspond by Ω into a horocycle.	(pg 464) "Definition- Corresponding Points"

45.



(From exercise) "Equidistant curves of $\vec{t}$ are the non-vertical rays in the upper half-plane emanating from 0."

Theorem 7.2: Poincare Disk- Angle of Parallelism

$$e^{-d} = \tan[\Pi(d)/2] \text{ where}$$

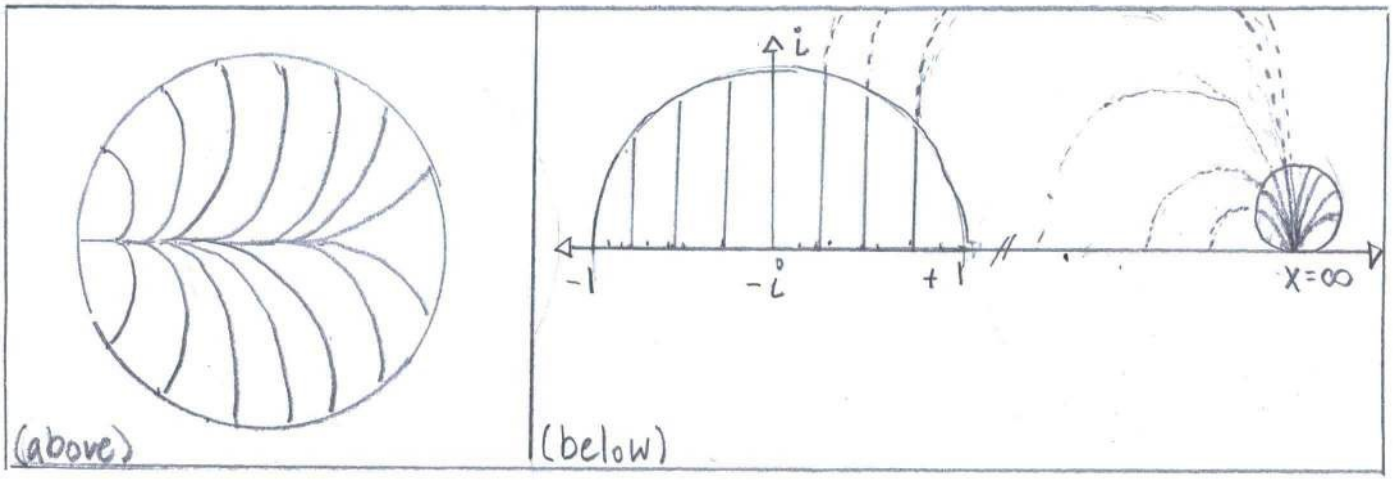
Π = Angle in radians

d = distance

$$\Pi(d) = \frac{\pi}{2} : d = 0 \text{ and } t$$

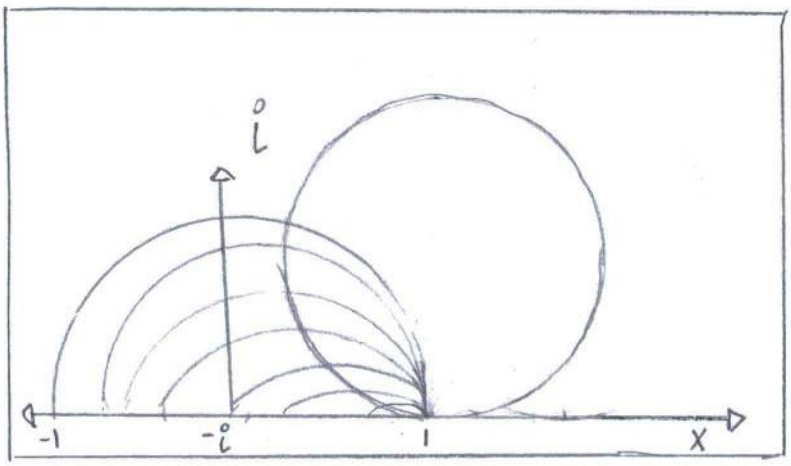
$$\Pi(d) \neq \frac{\pi}{2} : \boxed{0 \leq d \leq \infty}$$

46. (From exercise) "Show horocycles about ∞ are the horizontal lines in the upper half-plane."



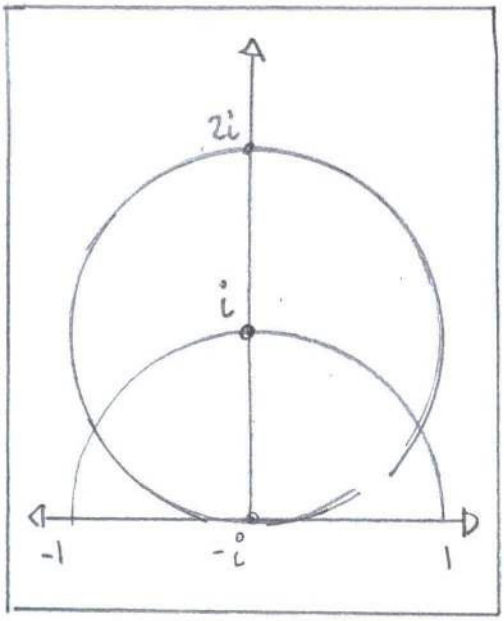
(From exercise) "Horocycles about X_0 are the circles in the upper half-plane tangent to the X-axis at X_0"

47. (From exercise) "Equidistant curves are intersections with the upper half-plane of circles cutting the X-axis in two points with centers of circles not lying on the X-axis"



"Poincare upper half-plane"

48. (From exercise) "Find the Euclidean radius and Euclidean center of the hyperbolic circle of hyperbolic center i passing through $2i$."



Proposition 7.5:

$r^2 = (\overline{oT})^2$ where $r = 1$

(original in book)

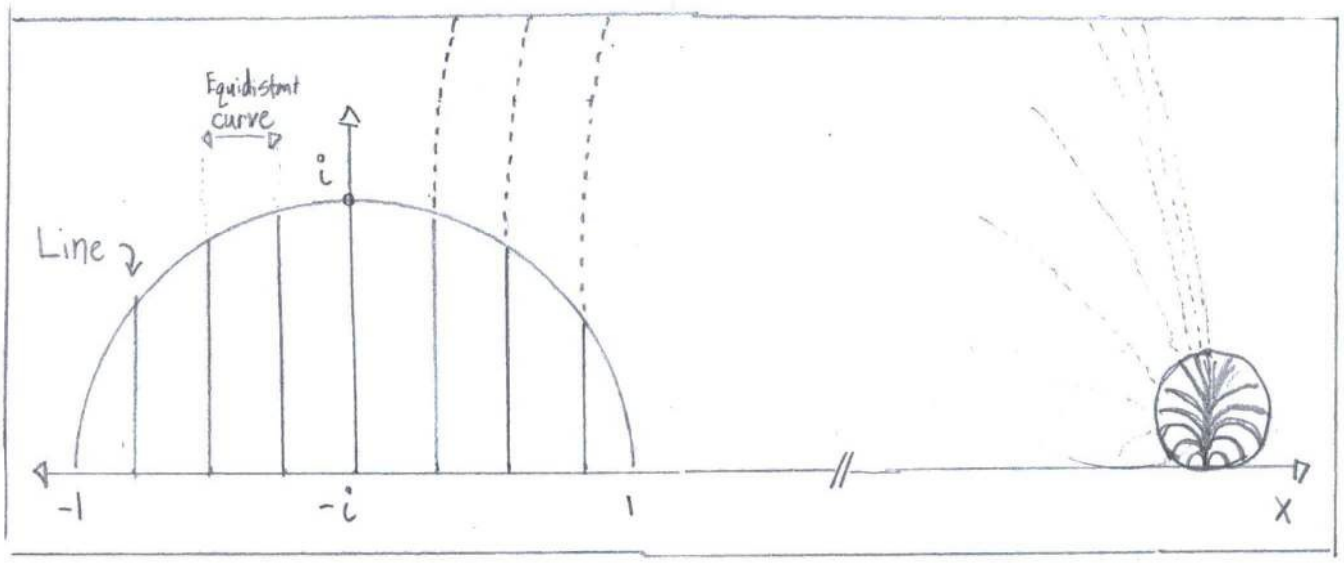
"r" = "radius"

$= \sqrt{x^2 + y^2}$

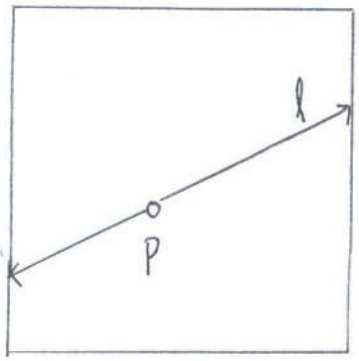
X = "center"

$= (x, 1 - x^2)$

49. (from exercise) "Use the Poincare upper half-plane model to demonstrate that in the hyperbolic plane, three points lie on a line, a circle, an equidistant curve, or a horocycle"



50.



(1) P lies on $l \Leftrightarrow (H_P R_l)^2 = I$

Proposition 9.25 (pg 444):

Motion	Formula	Determinant	Matrix
Reflection	$z \to \frac{az+b}{cz-a}$	$a^2+bc=1$	$\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$
Half-turn	$z \to \frac{\cos(\frac{\theta}{2})z - \sin(\frac{\theta}{2})}{(\sin\frac{\theta}{2})z + \cos(\frac{\theta}{2})}$	X	$\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$

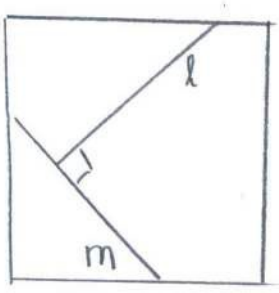
$$(H_P R_l)^2 = \left(\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$= \begin{bmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(2) $l \perp m \Leftrightarrow (R_l R_m)^2 = I$

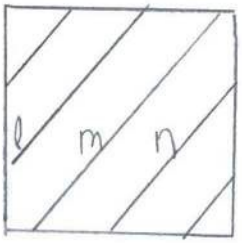
$$(R_l R_m)^2 = \left(\begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix}$$

$$= \begin{bmatrix} (a^2+bc)^2 & 0 \\ 0 & (a')^2+(b'c')^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(3) ℓ, m, n belong to a pencil $\Leftrightarrow (R_\ell R_m R_n)^2 = I$

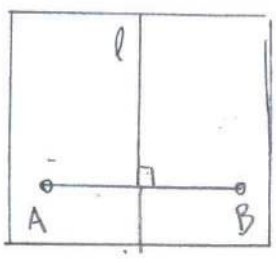
$$(R_\ell R_m R_n)^2 = \left(\begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix} \begin{bmatrix} a'' & b'' \\ c'' & -a'' \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \cdot \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix} \cdot \begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix} \cdot \begin{bmatrix} a'' & b'' \\ c'' & -a'' \end{bmatrix} \cdot \begin{bmatrix} a'' & b'' \\ c'' & -a'' \end{bmatrix}$$

$$= \begin{bmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{bmatrix} \begin{bmatrix} a'^2+b'c' & 0 \\ 0 & a'^2+b'c' \end{bmatrix} \begin{bmatrix} a''^2+b''c'' & 0 \\ 0 & a''^2+b''c'' \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(4) ℓ is the perpendicular bisector of AB

$\Leftrightarrow R_\ell H_b R_\ell H_A = I$

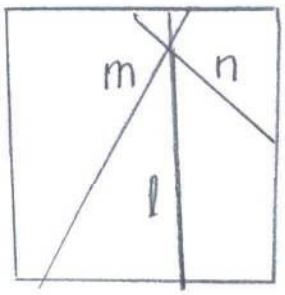
$$R_\ell H_B R_\ell H_a = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}\begin{bmatrix} a & b \\ c & -a \end{bmatrix}\begin{bmatrix} \cos'\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix}$$

where $\theta = \pi$

$$= \begin{bmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



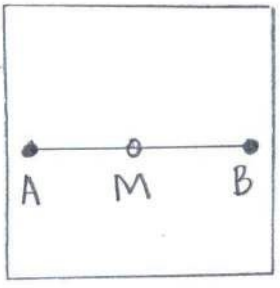
(5) Let m and n be intersecting lines. Then ℓ is a bisector of one of the angles formed by m and n. $\Leftrightarrow R_\ell R_m R_\ell R_n = I$

$$R_\ell R_m R_\ell R_n = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}\begin{bmatrix} a' & b' \\ c' & -a' \end{bmatrix}\begin{bmatrix} a & b \\ c & -a \end{bmatrix}\begin{bmatrix} a'' & b'' \\ -c'' & a'' \end{bmatrix}$$

$$=$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(6) M is the midpoint of $\overline{AB} \Leftrightarrow H_M H_B H_M H_A = I$

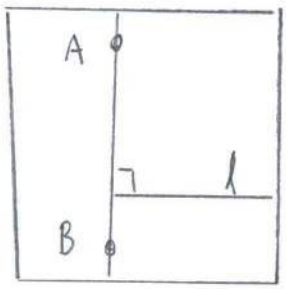
$$H_M H_B H_M H_A = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos'\frac{\theta}{2} & -\sin'\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

$$\times \begin{bmatrix} \cos''\frac{\theta}{2} & -\sin''\frac{\theta}{2} \\ \sin''\frac{\theta}{2} & \cos''\frac{\theta}{2} \end{bmatrix}$$

where $\theta = \pi$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



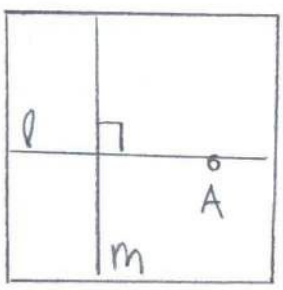
(7) $\overleftrightarrow{l} \perp \overleftrightarrow{AB} \Leftrightarrow (R_l H_A H_B)^2 = I$

$$(R_l H_A H_B)^2 = \left(\begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos''\frac{\theta}{2} & -\sin''\frac{\theta}{2} \\ \sin''\frac{\theta}{2} & \cos''\frac{\theta}{2} \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} a^2+b\cdot c & 0 \\ 0 & a^2+bc \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

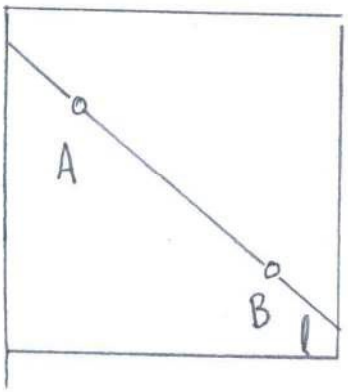


(8) A lies on a common perpendicular to both l and $m \Leftrightarrow (H_A R_l R_m)^2 = I$

$$(H_A R_l R_m)^2 = \left(\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & -b \\ c & a \end{bmatrix} \begin{bmatrix} a' & -b' \\ c' & a' \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(9) Given $A \neq B$, In Euclidean geometry,

$H_A R_\ell H_A = H_B R_\ell H_B \Leftrightarrow \overleftrightarrow{\ell} = \overleftrightarrow{AB}$ or $\overleftrightarrow{\ell} \parallel \overleftrightarrow{AB}$.

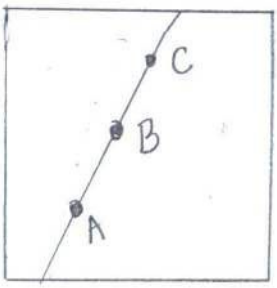
In hyperbolic geometry, the equation holds $\Leftrightarrow \overleftrightarrow{\ell} = \overleftrightarrow{AB}$.

$$H_A R_\ell H_A = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & -b \\ c & a \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & -b \\ c & a \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos'\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} a & -b \\ c & a \end{bmatrix} \begin{bmatrix} \cos'\frac{\theta}{2} & -\sin'\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix}$$

$$= H_B R_\ell H_B$$

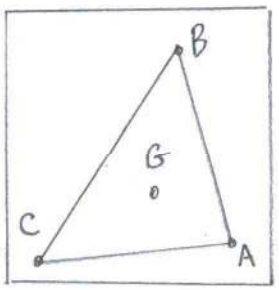


(10) In hyperbolic geometry, A, B, C are collinear $\Leftrightarrow (H_A H_B H_C)^2 = I$

$$(H_A H_B H_C)^2 = \left(\begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos'\frac{\theta}{2} & -\sin'\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos''\frac{\theta}{2} & -\sin''\frac{\theta}{2} \\ \sin''\frac{\theta}{2} & \cos''\frac{\theta}{2} \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$



(11) Assume A, B, C not collinear and that G is the centroid of $\triangle ABC$ (the point of intersection of medians). Then in Euclidean geometry,

$H_P H_C H_P H_B H_P H_A = I \iff P = G.$

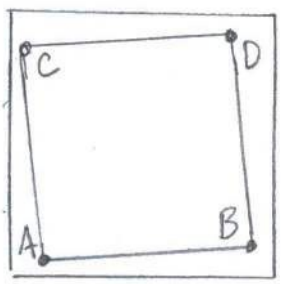
$$(H_P H_C H_P H_B H_P H_A) = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos'\frac{\theta}{2} & -\sin'\frac{\theta}{2} \\ \sin'\frac{\theta}{2} & \cos'\frac{\theta}{2} \end{bmatrix}$$

$$\times \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos''\frac{\theta}{2} & -\sin''\frac{\theta}{2} \\ \sin''\frac{\theta}{2} & \cos''\frac{\theta}{2} \end{bmatrix}$$

$$\times \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos'''\frac{\theta}{2} & -\sin'''\frac{\theta}{2} \\ \sin'''\frac{\theta}{2} & \cos'''\frac{\theta}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{...for half-turns at } \theta = \pi$$

$$= I$$



(12) Given $\square ABCD$ in the Euclidean plane. Then $H_A H_B H_C H_D = I \iff \square ABCD$ is a parallelogram

$$H_A \cdot H_B \cdot H_C \cdot H_D = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

$$\times \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

51. (From exercise) Determine whether each is invariant under the indicated group.

(i) $\mathcal{S}$ = all rotations about one given point, $\underline{\mathcal{G}}$ = all motions

If $\mathcal{S} \in \{S = R_m\}$ and $\underline{G} \in \{T = T_x, R_{0,0}, H_m, R_\ell, I\}$,

$$TST^{-1} = T_x R_{0,0} T_x^{-1}$$

$$\in \mathcal{S}$$

// So group $\mathcal{S}$ is not invariant to $\underline{G}$

(ii) $\mathcal{S}$ = all rotations about all points, $\underline{\mathcal{G}}$ = all motions

If $\mathcal{S} = \{S = R_x, R_y\}$ and $\underline{G} \in \{T = T_x, R_{0,0}, H_m, R_\ell, I\}$,

$$TST^{-1} = T_x \cdot R_{0,0} \cdot T^{-1}$$

$$= R_{-1/2,-1/2}$$

$$\in \mathcal{S}$$

Group $\mathcal{S}$ is invariant to group G.

(iii) $\mathcal{S}$ = all translations along one given line,
$\underline{G}$ = all rotations

If $\mathcal{S} \in \{S = T_x\}$ and $\underline{G} = \{T = T_x, R_{0,0}, H_m, R_\ell, I\}$,

$$T \cdot S \cdot T^{-1} = T_1 \cdot T_2 \circ T^{-1}$$
$$= T_2$$
$$\in \mathcal{S}$$

T_2 is in group $\mathcal{S}$ after an invariant operation. So $\mathcal{S}$ is invariant to group G.

(iv) $\mathcal{S}$ = all translations along lines, $\underline{G}$ = all motions

If $\mathcal{S} = \{S = T_x, T_y, T_c\}$ and $\underline{G} = \{T = T_x, R_{0,0}, H_{0,0}, R_m, I\}$,

$$T \cdot S \cdot T^{-1} = R \cdot T_2 \cdot R^{-1}$$
$$= T_2$$
$$\in \mathcal{S}$$

So, $\mathcal{S}$ is invariant to group $\underline{G}$.

(v) $\mathcal{S}$ = all motions, and $\underline{G}$ = all similarities

If $\mathcal{S} = \{S = T, R, H, R_m, G\}$ and $\underline{G} = \{T = D, R_{0,0}, T_x, R_x \cdot D,$

$R_x \cdot R_{0,0}, R_x \cdot T_x, I\}$

$$TST^{-1} = R_x \cdot D \cdot T_1 \cdot (R_x D)^{-1}$$

$$= D \cdot T_{-1}$$

$$\notin \mathcal{J}$$

After an invariant operation $(T \cdot ST^{-1})$, $D \cdot T_{-1}$ is not in group $\underline{G}$. Thus, $\mathcal{J}$ is not invariant to $\underline{G}$.

(vi) $\mathcal{J}$ = all dilations, $\underline{G}$ = all similarities

If $\mathcal{J} \in \{S = D_\gamma, D_\beta, D_\alpha, \ldots\}$ and

$\underline{G} \in \{T = D, R_{0,0}, T_x, R_x \cdot D, R_x \cdot R_{00}, R_x T_x, I\}$

$$T \cdot ST^{-1} = R_x \cdot D \cdot D \cdot (R_x \cdot D)^{-1}$$

$$= D$$

$$\cdot \notin \mathcal{J}$$

So, group $\mathcal{J}$ is not invariant to group $\underline{G}$

(vii) $\mathcal{J}$ = all parallel displacements, G = all similarities

If $\mathcal{J} = \{S = T_{x,n}\}$ and $\underline{G} = \{T = D, R_\infty, T_x, R_x \cdot D, R_x \cdot R_{00}, R_x \cdot T_x, I\}$

$$T \cdot S \cdot T^{-1} = R_x D \cdot T \cdot (R_x D)^{-1}$$
$$= I$$

So, group $\mathcal{S}$ is invariant to group $\mathcal{G}$.

A note, the above groups do not show an identity (1, I, or $\mathcal{S}$) in $\mathcal{S}$. While G does. Show an identity element.

(viii) $\mathcal{S}$ = all glides, $\mathcal{G}$ = all motions

If $\mathcal{S} = \{S = G, I\}$ and $\mathcal{G} = \{T = T_x, R_{0,0}, H_m, R_\ell, I\}$

$$T \cdot S \cdot T^{-1} = T \cdot G \cdot T^{-1}$$
$$= G$$
$$\in \mathcal{S}$$

So, group $\mathcal{S}$ is invariant to group $\mathcal{G}$.

(ix) $\mathcal{S}$ = all direct motions, $\mathcal{G}$ = all similarities

If. $\mathcal{S} = \{S = H_A, R_{0,0}, T, T_H, T_{II}, I\}$ and

$G = \{T = D, R_{0,0}, T_x, R_x \cdot D, R_x \cdot R_{0,0}, R_x \cdot T, I\}$

$$T \cdot S \cdot T^{-1} = (R_x D) T \cdot (R_x D)^{-1}$$
$$= T$$
$$\in \mathcal{S}$$

So, group $\mathcal{S}$ is invariant to $\underline{\mathcal{G}}$.

(X) $\mathcal{S}$ = all translations along $\overleftrightarrow{t}$, $\underline{\mathcal{G}}$ = all similarities having O as a fixed point.

IF $\mathcal{S} = \{S = T_t, I\}$ and $\underline{\mathcal{G}} = \{T = D_o, R_o, T_o, R_{xo} \cdot D_o, R_{xo} \cdot R_o, R_{xo} T_o, I\}$

$$T \cdot S \cdot T^{-1} = (R_x D) T_t (R_x D)^{-1}$$
$$= T_t$$
$$\in \mathcal{S}$$

Thus, group $\mathcal{S}$ is invariant to group $\underline{\mathcal{G}}$.

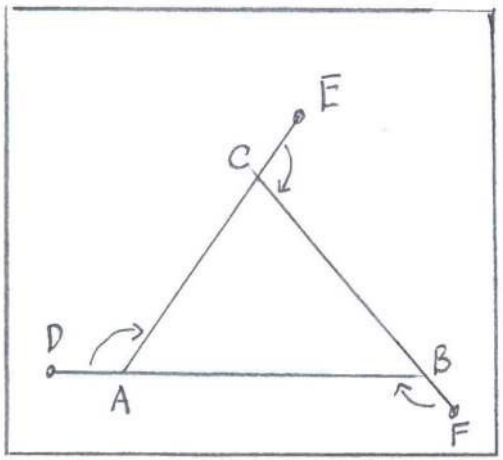
(Xi) $\mathcal{S}$ = all translations along t and $\underline{\mathcal{G}}$ = all similarities having t as an invariant line.

IF $\mathcal{S} = \{S = T_t, I\}$ and $G = \{S = D_t, R_t, T_t, R_{xt} \cdot D_t, R_{xt} \cdot R_t, R_{xt} \cdot T_t, I\}$

$$T \cdot S \cdot T^{-1} = (R_t D) T_t (R_t D)^{-1}$$
$$= T_t$$
$$\in \mathcal{S}$$

So group $\mathcal{F}$ is invariant to group $\underline{G}$.

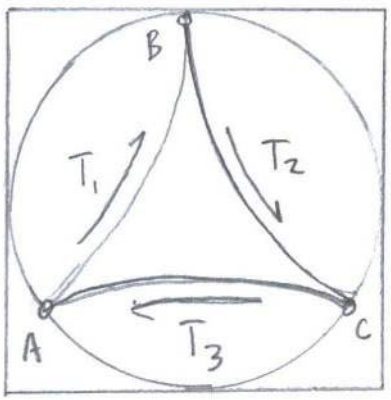
52.



"B. F. Thibaut"

Thibaut flawed in the proof. Triangles have an angle total at 180°.

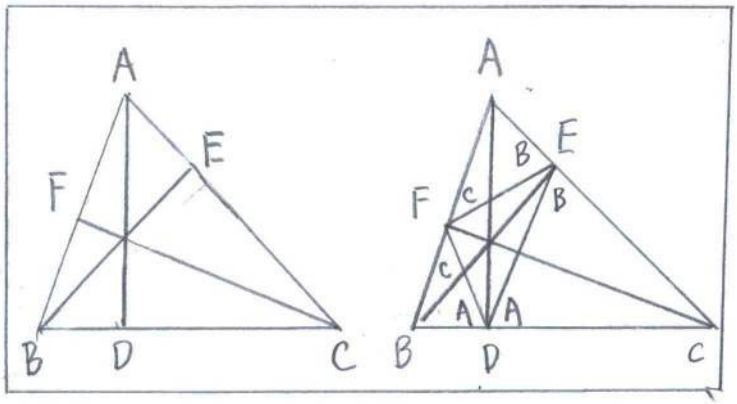
53.



(from exercise) $T_3T_2T_1$ is a rotation about A through $d°$, where $d°$ is a defect of ΔABC.

Proof: Hypothesis	Evidence
1) Triangle ΔABC	(pg 43) "Definition - Triangle.
2) Three points A, B, C are unique to an automorphism $T^{-1} \circ T = I$	Lemma 9.2
3) $T(\Delta ABC) = \Delta A'B'C'$	Corollary 1 to Proposition 9.2
4) $\measuredangle A \cong \measuredangle A'$, $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'}$	Proposition 9.5
5) $\Delta ABC \sim \Delta A'B'C'$	(pg 233) "Chapter 5 Exercise 12"
6) $T = T_3T_2T_1$	Actually, step #2

54.



(from exercise)

"The axis of G is the line $t = \overleftrightarrow{DF}$."

"Axis of a Euclidean glide around a triangle"

Proof: Hypothesis	Evidence
1) A Euclidean triangle $\triangle ABC$	(pg 43) "Definition-Triangle"
2) D, E, F are the feet to the perpendicular From A, B, C, considerably, where $a = \overline{BC}$, $b = \overline{AC}$, and $c = \overline{AB}$.	Proposition 4.4b
3) $\triangle ABE \sim \triangle ACF$ because $\measuredangle 1 = 90°$, $\measuredangle 2 = \measuredangle A$, and $\measuredangle 3 = 90° - \measuredangle A$.	(pg 215) "Definition-Similar Triangles"
4) $\triangle AEF \sim \triangle ABC$ since $\measuredangle 1 = A$, $\measuredangle 2 = \frac{\sin(90° - X)}{AF} = \frac{\sin(90° - Y)}{AE}$	(pg 215) "Definition-Similar Triangles" • Law of Sines

$= \frac{\cos(X)}{AF} = \frac{\cos(Y)}{AE}$

$\frac{AF}{b} = \frac{AE}{c}$

$AF = \frac{b}{c} AE = \frac{\cos(X)}{\cos(Y)} AE$

$AE = \frac{\sin(B)}{\sin(C)} AF$

$0 = \sin(C)\cos(X) - \cos(Y)\sin(B)$

$\measuredangle X = \measuredangle B$ and $\measuredangle Y = \measuredangle C$

- Horizontal shift
- Similar Triangles
- Sum and Difference
... interior angles are the largest triangle angles.

$\measuredangle 3 = 180° - \measuredangle 1 - \measuredangle 2$

5) $\triangle ABC \sim \triangle BFD$ because $\measuredangle 1 = \measuredangle B$, $\measuredangle 2 = \measuredangle C$ and $\measuredangle 3 = \measuredangle B$.

(pg 215) "Definition-Similar Triangles"

6) $\triangle ABC \sim \triangle CDE$ because $\measuredangle 1 = \measuredangle C$, $\measuredangle 2 = \measuredangle B$, and $\measuredangle 3 = \measuredangle A$

(pg 215) "Definition-similar triangles"

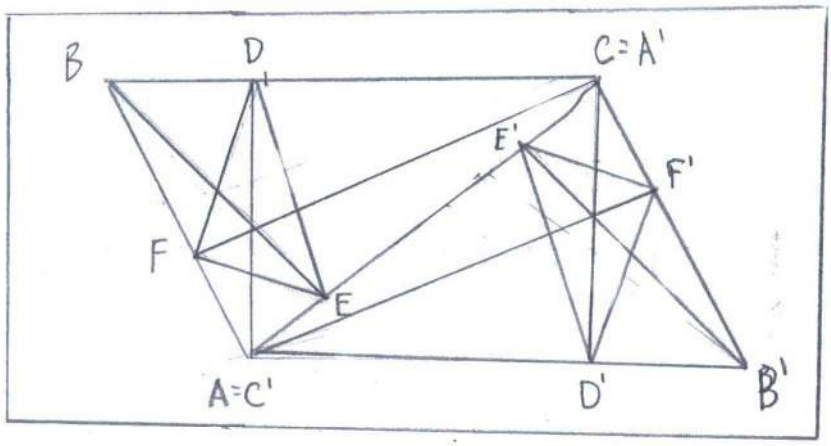
7) $\measuredangle AFE \cong \measuredangle BFD$, $\measuredangle AEF \cong \measuredangle CED$, $\measuredangle BDF \cong \measuredangle CDE$

(pg 215) "Definition-Similar triangles"

8) Line $\overleftrightarrow{t} = \overleftrightarrow{DF}$ is not a pencil to lines a, b, c, so axis $\overleftrightarrow{t}$ is a glide ($G = R_A R_B R_C$)

Proposition 9.19

55.

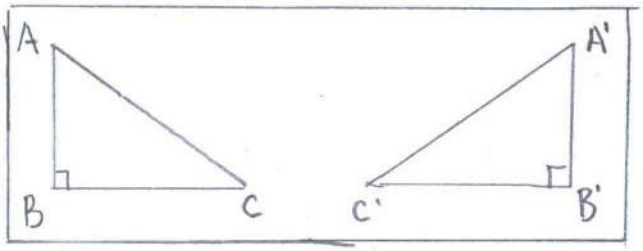


"Acute-angled glide"

(From exercise) "Show a translation in the direction of FD through a distance equal to the perimeter of $\triangle DEF$."

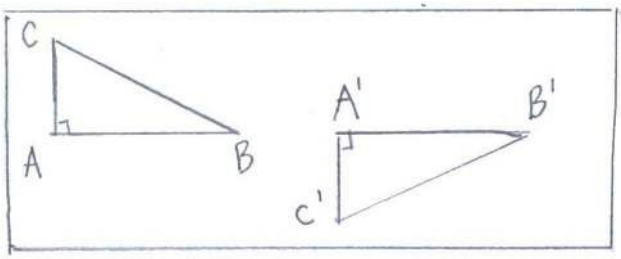
IF $G = T \cdot R_t$, then $T \cong \overline{DF} + \overline{FE} + \overline{DE}$.

56.



"Right angle at $\measuredangle B$"

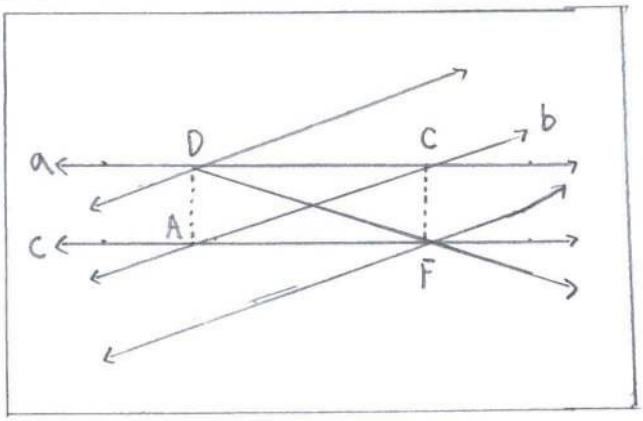
"$\overline{AB}$ is the axis to reflection in glide $G = T R_t$"



"Right angle not at $\measuredangle B$"

"$\overline{AB}$ is the axis to reflection in glide $G = T \cdot R_t$"

57.



"Euclidean glide"

(from exercise) "Given three lines a, b, c in the Euclidean plane such that a ⊥ c, b meets c at A, and b meets a at C"

(i) $R_a R_b R_c = R_a R_c R_b = R_b R_a R_c$

(ii) DF is the axis of glide $R_a R_b R_c$.

Proof:

Hypothesis	Evidence
1) Lines $\overleftrightarrow{a}, \overleftrightarrow{b}, \overleftrightarrow{c}$ are in a Euclidean plane where $\overleftrightarrow{a} \parallel \overleftrightarrow{c}$	Euclid's Postulate V
2) $\overrightarrow{b}$ intersects $\overleftrightarrow{c}$ at point A and $\overleftrightarrow{a}$ at point C.	Proposition 2.1
3) $G = R_a R_b R_c$	RAA hypothesis
Case (i) $b \perp (a, b)$:	
A) Line $\overrightarrow{b}$ is the only pencil to $\overrightarrow{a}, \overrightarrow{b}$	(pg 427) "Definition- Pencil of lines"
B) $\overrightarrow{b}$ is the axis of the glide	Proposition 9.19
C) G is the reflection	RAA result #1

about b.

Case(ii) $!_o(b \perp (a,c))$:

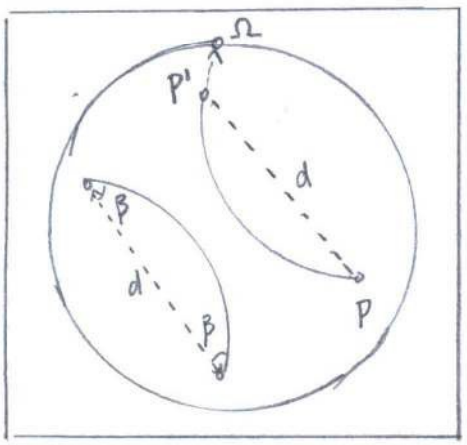
A) Between lines $\overleftrightarrow{a}$ and $\overleftrightarrow{c}$ are feet to the perpendiculars from A and C, specifically D and F. — Proposition 4.

B) At point D, $\overline{DF}$ is a pencil — (pg 427) "Definition-Pencil"

c) $G = R_a R_b R_c$
$= R_{b'} R_{c'}$
or
$= R_{a'} R_{b'}$ — RAA result #1

4) $G = R_a R_b R_c = R_c R_a R_b$ — RAA result #2

50.



"Translation through distance d"

Transformation $T_{\dots}$ sends $P_{\dots}$ the unique point P' on a ray $\overrightarrow{P\Omega}$ such that $PP' = d_{\circ}$

Lemma 7.4 - Poincare Congruent

$\overline{OB} = r(e^d - 1)/(e^d + 1)$

$\overline{PP'} = r(e^d - 1)/(e^d + 1)$

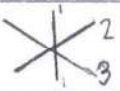
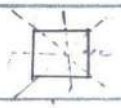
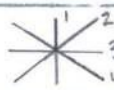
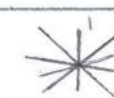
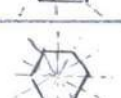
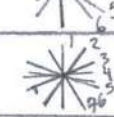
$$= r\left(\frac{1+\tan\beta}{1-\tan\beta} - 1\right) \Big/ \left(\frac{1+\tan\beta}{1-\tan\beta} + 1\right)$$

$$= r \cdot \beta$$

Transformation (T) is dependent on radius and either angle (β) or distance (d).

In the upper half-plane, Poincare length is $d = |\log(y/y')|$ where $x+iy$ translates to $x+iy'$. Transformation (T) would be $PP' = r \cdot \left|\frac{y-y'}{y+y'}\right| = r\left|\frac{y'-y}{y+y'}\right|$ and not collineation.

59. Dihedral Groups:

n	Image	Transformation		
		Ref•Rot	RefA•RefB	Relationship ($T = R_\alpha R_b$)
3	△			$T = R_\alpha R_b \quad \alpha, \beta = \{1,2,3\}$
4	□			$T = R_\alpha R_b \quad \alpha, \beta = \{1,\ldots,4\}$
5	⬠			$T = R_\alpha R_b \quad \alpha, \beta = \{1,\ldots,5\}$
6	⬡			$T = R_\alpha R_b \quad \alpha, \beta = \{1,\ldots,6\}$
7	⬡			$T = R_\alpha R_b \quad \alpha, \beta = \{1,\ldots,7\}$
...	...	...	...	...

60. Symmetry group of:

(i) ...LLLLLL... None

(ii) ...LΓLΓL... None

(iii) ...VVVVV... Reflection (R^{ef}_{n}), $\alpha = n^{th}$ letter

(iv) ...NNNNN... Rotation (R^{ot}_{n}), n^{th} letter

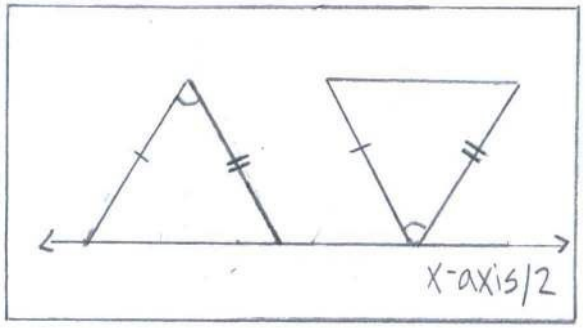
(v) ...V/\V/\V... Rotation (R^{ot}_{2n+1}) and Reflection (R^{ef}_{n})

(vi) ...DDDDD... None

(vii) ...HHHHH... Rotation (R^{ot}_{n}) and Reflection (R^{ef}_{n})

61. Groups (v) and (vii) were isomorphic, with a rotation and reflection.

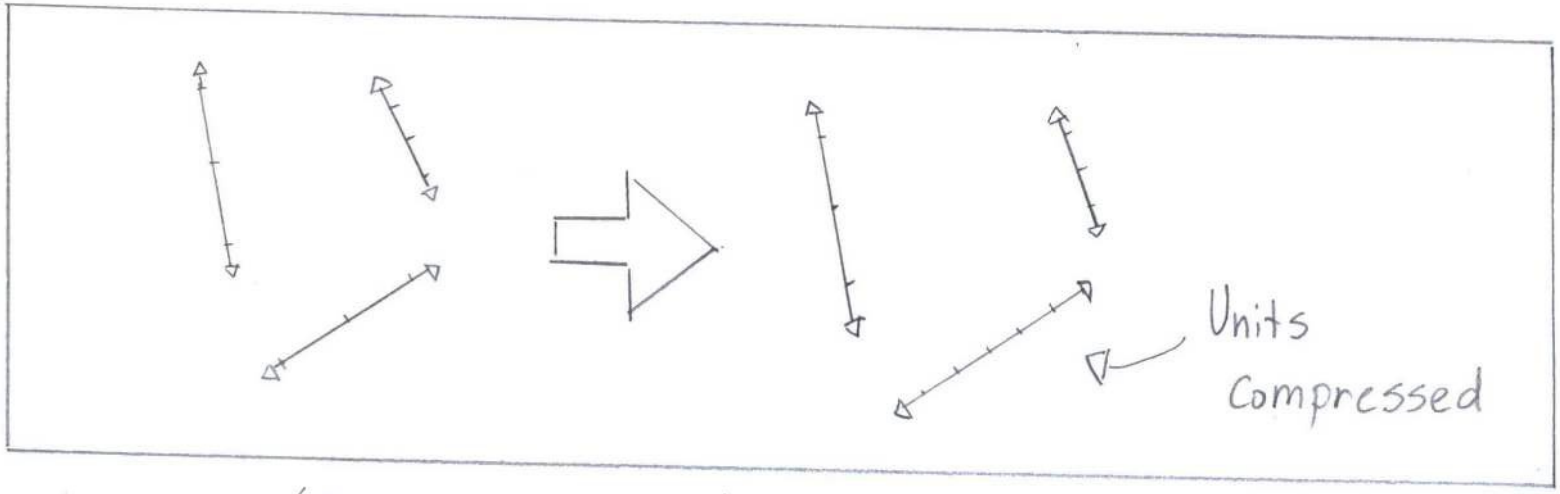
62.



Only automorphic similarities:

(pg 436) Similarities: Dilation, Rotation, Translation, and Reflection

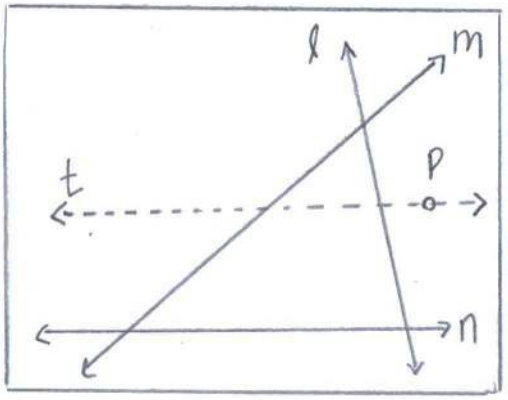
"Length has been changed" All similarities conserve the X-axis.



(from exercise) "The identity is the only automorphism of this model"

Proof: Hypothesis	Evidence
1) Three non-currant lines	(pg 20) "Definition - Units"
2) Transformation T compressed units	(pg 408) "Definition - Motion"
3) Identity is the automorphism	Proposition 9.1

64.



"Incident Plane"

(from exercise) "A model of both incidence and betweeness axioms is automatically two-dimensional."

Proof: Hypothesis	Evidence
1) Lines $\overleftrightarrow{l}, \overleftrightarrow{m}$, and $\overleftrightarrow{n}$ form a triangle	(pg 43) "Definition-Triangle"
2) Point P is exterior	(pg 117) "Definition-Exterior"
3) Line $\overleftrightarrow{t}$ intersects $\overleftrightarrow{l}$ and $\overleftrightarrow{m}$ at O and P, for P*O*Q	(pg 114) "Pasch's Theorem"
4) Pasch's theorem requires Axioms B-1, B-2, B-3 and Axioms I-1, I-2, While, I-3 is line $\overleftrightarrow{t}$ because two intersections at most.	RAA hypothesis → result

65. (From exercise) Analogue of Proposition 9.16(c)

"For any two triples of points on the projective line $\mathcal{P}^1(K)$, there is a unique projective transformation mapping one triple to another"

Cite: Ewald, G.
Geometry: an introduction
Johannes Gutenberg-Universität Mainz

If (y_0, y_1, y_2) be a triple, then (y_0', y_1', y_2') a projective transformation from (y_0, y_1, y_2).

Equations formed by Projection:
$$y_{n+1} = a_0 y_0 + a_1 y_1 + a_2 y_2$$
$$y'_{n+1} = a_0 y_0 + a_1' y_1' + a_2' y_2'$$

Transformation (T): $\lambda_i y_i' = y_i \cdot T$

Steps to Transformation (T):

① Eigenvalues:
$$\lambda_{n+1} y'_{n+1} = \lambda_{n+1}(a_0' y_0' + a_1' y_1' + a_2' y_2')$$
$$y_{n+1} T = (a_0 y_0 + a_1 y_1 + a_2 y_2) T$$
$$= a_0 \lambda_0 y_0' + a_1 \lambda_1 y_1' + a_2 \lambda_2 y_2'$$

② Coefficients: $a'_n = \frac{\lambda_n}{\lambda_{n+1}} a_n$

③ Matrix: $\begin{pmatrix} y_0 \\ y_1 \\ y_2 \end{pmatrix} T = \begin{pmatrix} \lambda_0 y_0' \\ \lambda_1 y_1' \\ \lambda_2 y_2' \end{pmatrix}$

$$= \lambda_{n+1} \begin{pmatrix} \frac{a_0'}{a_0} y_0 \\ \frac{a_1'}{a_1} y_1 \\ \frac{a_2'}{a_2} y_2 \end{pmatrix}$$

④ Solution: $T = \lambda_{n+1} (y_0\ y_1\ y_2) \cdot \begin{pmatrix} \frac{a_0'}{a_0} y_0 \\ \frac{a_1'}{a_1} y_1 \\ \frac{a_2'}{a_2} y_2 \end{pmatrix}$

66. (from exercise) "The cross-ratio is preserved: $(P_1 P_2, P_3 P_4) = (P_1' P_2', P_3' P_4')$

Proof:

Hypothesis	Evidence
1) $\frac{\begin{vmatrix} s_1 & t_1 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_2 & t_2 \\ s_4 & t_4 \end{vmatrix}}{\begin{vmatrix} s_2 & t_2 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_1 & t_1 \\ s_4 & t_4 \end{vmatrix}} = (\overline{P_1 P_2}, \overline{P_3 P_4})$	(Pg 470) "Definition-cross-ratio"
2) M is a matrix	RAA hypothesis

3) $(\overline{P_1P_2}, \overline{P_3P_4}) \cdot M$

$$= \frac{\begin{vmatrix} s_1 & t_1 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_2 & t_2 \\ s_4 & t_4 \end{vmatrix}}{\begin{vmatrix} s_2 & t_2 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_1 & t_1 \\ s_4 & t_4 \end{vmatrix}} \cdot M$$

$$= (\overline{P_1'P_2'}, \overline{P_3'P_4'})$$

RAA result

67. (pg 470) "Cross-ratio"

$$\frac{\begin{vmatrix} s_1 & t_1 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_2 & t_2 \\ s_4 & t_4 \end{vmatrix}}{\begin{vmatrix} s_2 & t_2 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_1 & t_1 \\ s_4 & t_4 \end{vmatrix}} = (\overline{P_1P_2}, \overline{P_3P_4})$$

(pg 319) "Cross-ratio"

$$\frac{(\overline{AP})(\overline{BQ})}{(\overline{BP})(\overline{AQ})} = (\overline{AB}, \overline{PQ})$$

(pg 368) "Signed cross-ratio"

$$\frac{(\underline{AC}/\underline{CB})}{(\underline{AD}/\underline{DB})} = (\underline{AB}, \underline{CD})$$

.. where $\underline{AB} = -\underline{BA}$

The formula above relate!

$$(\overline{P_1P_2}, \overline{P_3P_4}) = \frac{\begin{vmatrix} s_1 & t_1 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_2 & t_2 \\ s_4 & t_4 \end{vmatrix}}{\begin{vmatrix} s_2 & t_2 \\ s_3 & t_3 \end{vmatrix} \begin{vmatrix} s_1 & t_1 \\ s_4 & t_4 \end{vmatrix}}$$

$$= \frac{\begin{vmatrix} -s_1 & -t_1 \\ -s_3 & -t_3 \end{vmatrix} \begin{vmatrix} -s_2 & -t_2 \\ -s_4 & -t_4 \end{vmatrix}}{\begin{vmatrix} -s_2 & -t_2 \\ -s_3 & -t_3 \end{vmatrix} \begin{vmatrix} -s_1 & -t_1 \\ -s_4 & -t_4 \end{vmatrix}}$$

$$= (\underline{P_1P_2}, \underline{P_3P_4})$$

68. (from exercise) Deduce from Exercises 65 and 66 that if (P_1, P_2, P_3) and (P_1', P_2', P_3') are any two triplets of points on $P'(K)$, if T is the unique projective transformation carrying the first triple onto the second, and if P_4 is any fourth point, then the image P_4' of P_4 under T is uniquely determined by the equation $(\overline{P_1P_2}, \overline{P_3P_4}) = (\overline{P_1'P_2'}, \overline{P_3'P_4'})$.

RAA deduction: (P_1, P_2, P_3) projects to (P_1', P_2', P_3'), then P_4 to P_4'.*

*see #65, 66

69. (from exercise) "$T(x,y) = (\sigma x, \sigma y)$ defines an automorphism of Pythagorean Plane F^2 and that if $A=(0,0)$, $B=(1,0)$, and $C=(1,t)$, then $\measuredangle CAB$ is not mapped onto a congruent angle by T."

Proof: Hypothesis	Evidence
1) σ fixes $\mathbb{R}$ and sends t to $t+1$	RAA hypothesis
2) Points $A=(0,0)$, $B=(1,0)$, and $C=(1,t)$	(pg 12) "Definition-Points"
3) $A' = T(A) = (\sigma\cdot 0, \sigma\cdot 0) = (0,0)$ $B' = T(B) = (\sigma\cdot 1, \sigma\cdot 0) = (1,0)$ $C' = T(C) = (\sigma\cdot 1, \sigma\cdot t) = (1, t+1)$	(pg 408) "Definition-Transform"
4) $\measuredangle CAB \neq \measuredangle C'A'B'$ because $\theta = \cos^{-1}\left(\frac{1}{\sqrt{t^2+1}}\right)$ $= \cos^{-1}\left(\frac{1}{\sqrt{1+(t+1)^2}}\right)$	RAA result

170.

(pg 420) Proposition 9.15

"In a Euclidean plane, the product $H_A H_B H_C$ of three half-turns is a half-turn. In a hyperbolic plane, the product is a half-turn only when A, B, C are collinear; if they are not, the product can be a rotation, a translation, or a parallel displacement.

Hyperbolic cases (when A, B, C are not collinear)

1) $H_A H_B H_C$ is a rotation that is not a half-turn iff A, B, C lie on a circle.

Proof:

Hypothesis	Evidence
1) A circle has radius $OA = OB = OC$	Euclid's Postulate V
2) $H_A H_B H_C = R_0 \cdot R_0 \cdot R_0$ $= (R_0)^2 R_0$ $= I \cdot R_0$	Direct proof.

$= R_0$

$= H_0$

2) A translation iff A, B, C lie on an equidistant curve."

Proof: Hypothesis	Evidence
1) Points A, B, C lie on an equidistant curve in a Poincare disc	(pg 197) "Definition - Equidistant" -
2) $H_A H_B H_C = R_0 \circ R_0 \circ R_0$ $= (R_0)^2 \circ R_0$ $= I \circ R_0$ $= R_0$ $= H_0$	Direct Proof

3) A parallel displacement iff A, B, C lie on a horocycle. on a curve

Proof: Hypothesis	Evidence
1) Points A, B, C lie on	(pg 197) "Definition -

a horocycle.

2) $H_A H_B H_C = R_0 \cdot R_0 \cdot R_0$

$= (R_0)^2 R_0$

$= I \cdot R_0$

$= H_0$

Equidistant"

Direct Proof

71. Table of Motions:

	Orientation	Points fixed	Invariant Lines	Ideal fixed points (hyperbolic plane)
Identity	Direct	All	All	All
Reflection (R_ℓ)	Opposite	Points on ℓ	ℓ and all $m \perp \ell$	Two ends of ℓ
Half-turn (H_A)	Direct	A	Lines on A	None
Rotation about A, not involutory	Direct	A	None	None
Euclidean translation on t	Direct	None	t and all $u \parallel t$	–
Hyperbolic translation on t	Direct	None	None	Ω
Parallel displacement (Ω)	Direct	None	None	Ω
Glide along t.	Opposite	None	t	The two ends of ℓ

- Invariant cycles

Identity	Reflection	Half-turn	Rotation	Euclidean Translation	Hyperbolic Translation	Parallel displacement	Glide on t
All	Circle Horocycle	Circle Equidistant curve	Circle Horocycle Equidistant curve	None	Equidistant curve	Equidistant curve	circle.

72. (from exercise) "Every circle in $\mathbb{R}^2$ passes through I and J"

Proof: Hypothesis	Evidence
1) Isaac and Jacob $I=[1,i,0]$ $J=[1,-i,0]$	(pg 471) "Definition- Isaac and Jacob"
2) All circles in a projective plane satisfy $x^2+y^2=0$ because size at the horizon nearly a point, in addition to circles (when $r\cong 0$).	RAA hypothesis
3) Solutions to $x^2+y^2=0$ are Isaac or Jacob.	RAA result.

(from exercise) "If a conic in $\mathbb{R}^2$ passes through I and J, then it is a circle."

Proof:

Hypothesis	Evidence
1) Conics have a form $x^2-y^2=0$ or $y^2-x^2=0$ at infinity	RAA hypothesis
2) By I or J, the conic becomes a formula closer to a circle $x^2+y^2=0$. e.g. conics become circles from the definition circles at infinity.	Formula: Parabola $y=a(x-h)^2+k$ Hyperbola $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$ or $\frac{(y-k)^2}{b^2}-\frac{(x-h)^2}{a^2}=1$ Circle $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=r^2$

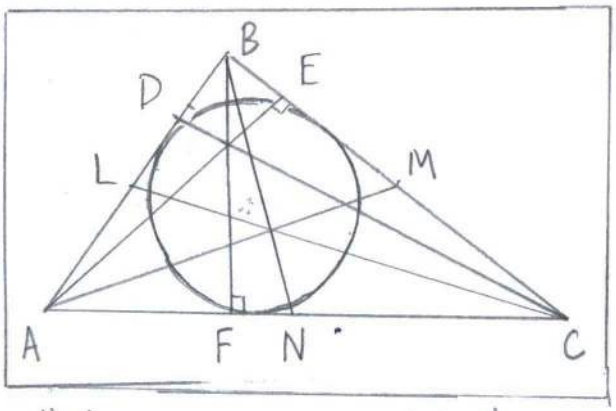
RAA result

(from exercise) "An affine transformation of $\mathbb{R}^2$ is a similarity if and only if, when extended to the complex projective plane

in the natural way, it leaves the set invariant $\{I, J\}$."

Proof: Hypothesis	Evidence
1) I and J in an affine plane $I=[1,i]$, and $J=[1,-i]$	RAA conjecture
2) Circles map to circles by I and J, in the form $[x,y]$ and through the equation $x^2+y^2=0$	Theorem 9.3
3) The affine transform. of $\mathbb{R}^2$ is a Similarity	RAA conclusion

73.



"Nine-point circle"

(pg 399) Harmonic Conjugate

"A, B, C, D are four collinear points in a Euclidean plane such that $(\overline{AB}, \overline{CD})=1$. Then we say C and D are harmonic conjugates

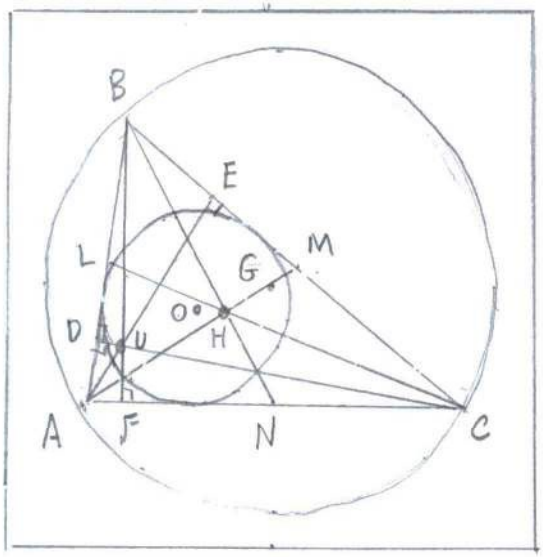
(from exercise) "The center V of the nine-point circle is the harmonic conjugate of the circumcenter O with respect to the orthocenter H and the centroid G"

$$(HG, VO) = \frac{\overline{HV}}{\overline{HO}} \cdot \frac{\overline{GO}}{\overline{GV}}$$

$$= \frac{(\overline{HO}/2)}{\overline{HO}} \cdot \frac{(\overline{HG}/2)}{(\overline{HG}/4)}$$

$$= 1$$

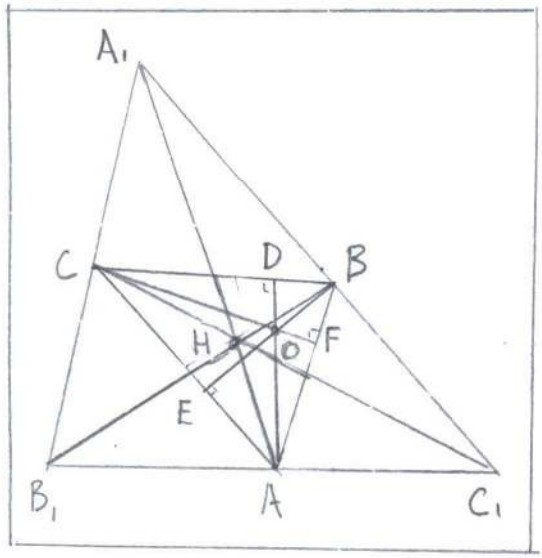
74.



"Centers of Similitude"

A cross-ratio at -2 implies $\frac{\overline{HU}}{\overline{HO}}\frac{\overline{GO}}{\overline{GU}} = -2$ from a dilation T'. The circumcenter encompasses the triangle in a nine-point circle.

75.

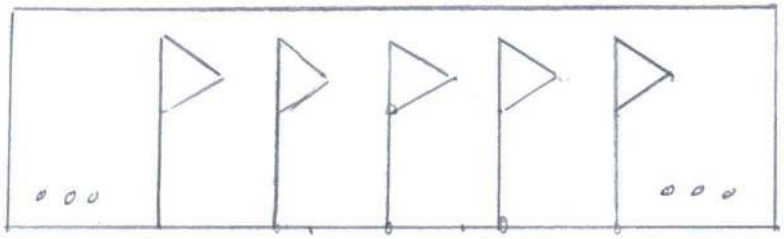


"circumcenter"

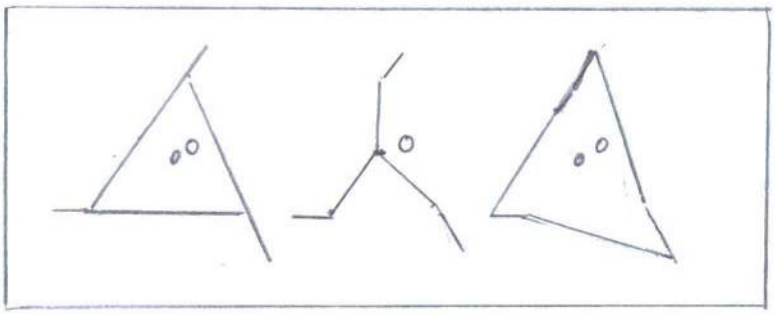
(from exercise) "The distance from the circumcenter O of a triangle to a side is half the distance from the orthocenter H to the opposite vertex."

Dilation T=-2 maps $\triangle ABC$ to $\triangle A'B'C'$ with parallel sides and a medial triangle. Wildly, $\overline{OD} = \frac{1}{2}\overline{HA}$ (2cm = $\frac{1}{2}$ 4cm), $\overline{OF} = \frac{1}{2}\overline{HC}$ (1cm = $\frac{1}{2}$ 2cm), and $\overline{OE} = \frac{1}{2}\overline{HB}$ (1cm = $\frac{1}{2}$ 2cm). Very close by hands.

76.



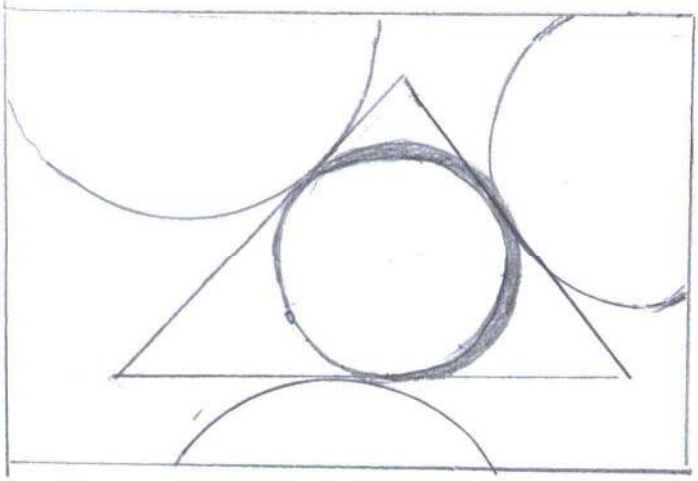
"Infinite cyclic group of symmetry"



"Symmetry groups"

Three dilemma resided in symmetries. The first problem was a minimal set of generators. e.g the smallest group with symmetry. A second problem became relationships between high-order operations and generators. Lastly, the structure in the group wasn't familiar.

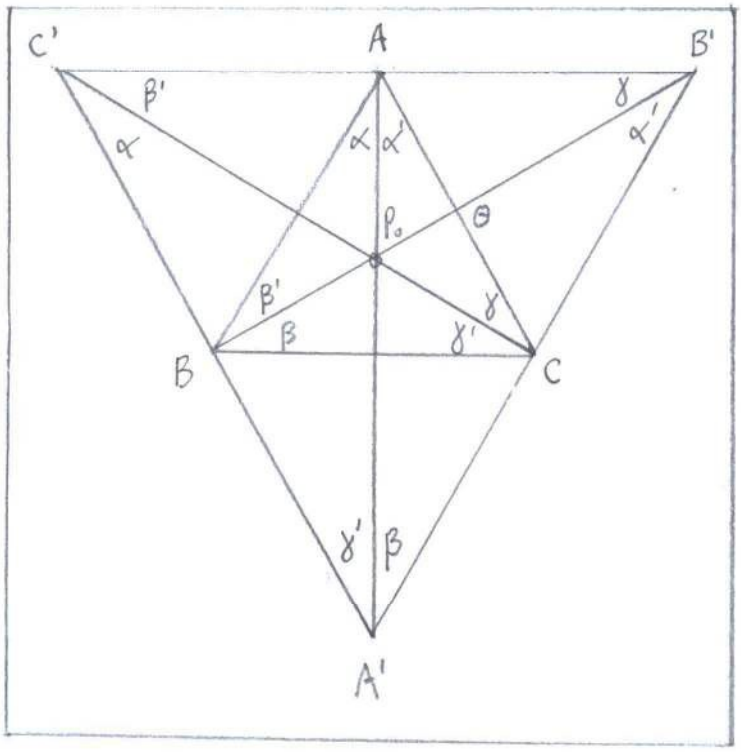
77.



"Feuerbach Point."

The nine-point circle is tangent to the inscribed circle and three escribed circles. Supposedly, no physical object exists for Feuerbach.

78. (From exercise) "There is a unique point P_o which the function $d(P) = \overline{PA} + \overline{PB} + \overline{PC}$ achieves a minimum value... that P_o lies in the interior of $\triangle ABC$... that it can be constructed with a straight-edge and compass... that $120^\circ = (\measuredangle AP_oB)^\circ = (\measuredangle BP_oC)^\circ = (\measuredangle CP_oA)^\circ$ ... and that $d(P_o)^2 = a^2 + b^2 - 2ab\cos(C + \pi/3)$ where $a = \overline{BC}$, $b = \overline{AC}$.



A solution!